

POSITIVE SECTIONAL CURVATURE ON ALL SMOOTH 7-SPHERES

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ABSTRACT. We construct a smooth Riemannian metric with strictly positive sectional curvature on every smooth homotopy seven-sphere. For each smooth structure, we construct positively curved metrics on two seven-dimensional disks whose induced boundary metrics agree under the prescribed attaching map. Under this identification, a gluing theorem then yields the required smooth metric on the closed manifold. The construction applies to all 28 oriented diffeomorphism classes.

1. INTRODUCTION

Determining which smooth manifolds admit metrics of positive sectional curvature is a fundamental problem in Riemannian geometry. On manifolds homeomorphic to spheres, this question brings the distinction between topology and differentiable structure into sharp focus. In 1956, Milnor constructed smooth seven-dimensional manifolds homeomorphic, but not diffeomorphic, to the standard sphere S^7 [10].

A smooth homotopy seven-sphere is a closed smooth manifold homotopy equivalent to S^7 . Its oriented diffeomorphism class lies in the group Θ_7 under connected sum. Kervaire and Milnor proved that

$$\Theta_7 \cong \mathbb{Z}/28\mathbb{Z},$$

with the standard sphere representing the identity [9]. A smooth homotopy seven-sphere is called *exotic* when it is not diffeomorphic to the standard sphere.

Since the standard sphere carries a metric of constant positive sectional curvature, the existence of exotic spheres raises a natural question:

Which smooth structures on S^7 admit metrics of positive sectional curvature?

This question can be asked separately in each of the 28 oriented classes.

Let g be a Riemannian metric on a smooth manifold M , and write $\langle \cdot, \cdot \rangle$ for its inner product. For linearly independent vectors $u, v \in T_p M$, the sectional curvature is

$$\sec_p(u, v) = \frac{\langle R(u, v)v, u \rangle}{\langle u, u \rangle \langle v, v \rangle - \langle u, v \rangle^2},$$

where R is the Riemann curvature tensor with the sign convention specified below. Let $\text{Gr}_2(T_p M)$ denote the Grassmannian of two-planes in $T_p M$. We also write $\sec_p(\Pi)$, or $\sec_g(\Pi)$ to emphasize the metric. Positive sectional curvature means that $\sec_p(\Pi) > 0$ for every $p \in M$ and every $\Pi \in \text{Gr}_2(T_p M)$.

For a positively curved metric and $0 < \delta < 1$, strict δ -pinching means

$$\delta < \sec_{\min} / \sec_{\max}.$$

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For pointwise pinching, the extrema are taken over $\text{Gr}_2(T_p M)$ separately at each point p . On a compact manifold, global pinching uses the extrema over all points and two-planes. The differentiable sphere theorem of Brendle and Schoen [1] states that a closed, simply connected Riemannian manifold of dimension at least four is diffeomorphic to a sphere if it is strictly pointwise $1/4$ -pinched, that is,

$$\sec_{\min}(p) > \frac{1}{4} \sec_{\max}(p) > 0 \quad \text{at every point } p.$$

An exotic seven-sphere therefore admits no strictly pointwise $1/4$ -pinched metric. The existence of positive curvature without this pinching condition is a different question.

Gromoll and Meyer constructed the first nonnegatively curved exotic sphere in 1974 [6]; we denote it by Σ_{GM} . Their description as a biquotient of $\text{Sp}(2)$ made Lie group geometry and Riemannian submersions available for its study. Grove and Ziller [7] subsequently constructed nonnegatively curved metrics on all Milnor spheres, i.e., the homotopy seven-spheres that arise as S^3 -bundles over S^4 . Goette, Kerin and Shankar [5] extended this existence result to every exotic seven-sphere.

For the Gromoll–Meyer sphere, Wilhelm [16] obtained positive sectional curvature almost everywhere, and Eschenburg and Kerin [4] constructed an almost-positively curved metric with an explicitly described exceptional set. Here, almost positive curvature means nonnegative sectional curvature everywhere and positive curvature on every two-plane over an open dense set of points. Strictly positive constructions on this sphere have been proposed by Petersen and Wilhelm [13], Ouyang [12], and Guo, Fang and Lu [8].

1.1. Main results. Our main theorem is the following.

Theorem 1.1. *Every smooth homotopy seven-sphere admits a smooth Riemannian metric with strictly positive sectional curvature. Equivalently, for each $[\Sigma] \in \Theta_7$, there are a smooth Riemannian metric g and a constant $c_\Sigma > 0$ such that, for every $p \in \Sigma$,*

$$(1.1) \quad \sec_g(\Pi) \geq c_\Sigma \quad (\Pi \in \text{Gr}_2(T_p \Sigma)).$$

Remark 1.2. The uniform lower bound follows from compactness of the Grassmann bundle.

To prove Theorem 1.1, we use the two-disk models of Durán, Püttmann and Rigas [3]. For each $n \in \mathbb{Z}$, the model Σ_n represents $n \bmod 28$ in Θ_7 and is obtained by attaching two seven-disks D_N, D_S along a map $\phi_n : \partial D_N \rightarrow \partial D_S$. In the polar boundary coordinates, ϕ_n is represented by σ^n for a fixed diffeomorphism $\sigma : S^6 \rightarrow S^6$. Section 2 specifies this map and the orientation convention. We call the disks *marked* once these boundary coordinates and the attaching map are fixed. The following theorem constructs positive metrics compatible with this prescribed attaching map.

Theorem 1.3 (Compatible positive disk metrics). *For every $n \in \mathbb{Z}$, the marked disks D_N, D_S defining Σ_n admit smooth Riemannian metrics g_N, g_S of strictly positive sectional curvature, including at their boundaries. Let h_N, h_S be the induced boundary metrics and B_N, B_S the outward second fundamental forms, with convention (1.2). The marked boundary identification $\phi_n : \partial D_N \rightarrow \partial D_S$, represented by σ^n in the boundary coordinates extending over the disks, satisfies*

$$h_N = \phi_n^* h_S, \quad B_N + \phi_n^* B_S > 0.$$

The last inequality signifies positive definiteness on the common boundary tangent bundle.

Remark 1.4. Brendle–Schoen’s sphere theorem says that sufficiently pinched positive sectional curvature forces a simply connected closed manifold to be diffeomorphic to the standard sphere. In particular, the condition

$$K_{\min}(p) > \frac{1}{4}K_{\max}(p) > 0$$

excludes exotic spheres. Our main theorem would establish the complementary statement:

$$\boxed{\text{Positive sectional curvature alone excludes none of the smooth types on } S^7.}$$

Thus, the quantitative distribution of curvature, rather than its positive sign alone, becomes essential to this smooth rigidity phenomenon.

1.2. The construction. Fix $n \in \mathbb{Z}$. [Section 2](#) constructs a principal right S^3 -bundle $\pi_n : P_n \rightarrow S^7$, with right action $R_h(p) = ph$, and a smooth free left action

$$\star : S^3 \times P_n \longrightarrow P_n, \quad (q, p) \longmapsto q \star p, \quad q \star (ph) = (q \star p)h.$$

The quotient by this second action is the orbit space

$$\Sigma_n := P_n / \sim_\star, \quad p \sim_\star p' \iff p' = q \star p \text{ for some } q \in S^3,$$

with quotient map $\varpi_n : P_n \rightarrow \Sigma_n$, $p \mapsto [p]_\star = \{q \star p : q \in S^3\}$. We also write $P_n / S_\star^3$ for this orbit space and call it the *star quotient* of P_n .

Choose complementary closed polar disks $U_N, U_S \subset S^7$ with common boundary. Their preimages are invariant under $\star$. For $\alpha \in \{N, S\}$, set

$$P_\alpha := \pi_n^{-1}(U_\alpha), \quad D_\alpha := P_\alpha / S_\star^3, \quad \varpi_\alpha : P_\alpha \longrightarrow D_\alpha, \quad p \longmapsto [p]_\star.$$

In the polar trivializations $P_\alpha \cong D^7 \times S^3$, where D^7 is the closed unit disk in $\text{Im } \mathbb{H} \oplus \mathbb{H}$, the actions are

$$(z, u)h = (z, uh), \quad q \star (z, u) = (qzq^{-1}, qu),$$

where conjugation acts on both components of z . The maps $[(z, u)]_\star \mapsto u^{-1}zu$ identify D_N, D_S smoothly with seven-disks. In their boundary coordinates, the attaching map $\phi_n : \partial D_N \rightarrow \partial D_S$ is $y \mapsto \sigma^n(y)$, and

$$\Sigma_n \cong D_N \cup_{\phi_n} D_S.$$

Thus the metrics to be glued are on D_N, D_S ; we construct them as quotients of metrics on the ten-dimensional spaces P_N, P_S .

For a fixed $\alpha \in \{N, S\}$, write $B = U_\alpha$, $P = P_\alpha$, $D = D_\alpha$, $\pi = \pi_n|_{P_\alpha}$, and $\varpi = \varpi_\alpha$. Choose a base metric g_B , a smooth principal connection $\omega \in \Omega^1(P; \mathfrak{sp}(1))$, and a smooth positive function $r : B \rightarrow (0, \infty)$. The connection specifies the splitting

$$TP = H \oplus V, \quad H = \ker \omega, \quad V = \ker d\pi.$$

Let Q be the Ad-invariant inner product on $\mathfrak{sp}(1)$ for which S^3 has curvature one. The metric

$$G = \pi^* g_B + (r \circ \pi)^2 Q(\omega, \omega)$$

makes H orthogonal to V , lifts g_B to H , and gives the fibre over b the round metric of radius $r(b)$. We call G the *connection metric with fibre radius r* .

We choose the data so that the $\star$ -action preserves G . Put

$$\mathcal{H}_p^\star := \left(T_p(S^3 \star p) \right)^{\perp_G} = (\ker d\varpi_p)^{\perp_G}.$$

These are the *star-horizontal* spaces; they are generally different from the principal-horizontal spaces H_p . The quotient metric g on D is defined by

$$g_{[p]\star}(d\varpi_p X, d\varpi_p Y) = G_p(X, Y), \quad X, Y \in \mathcal{H}_p^\star.$$

Invariance makes this definition independent of the representative p , and $\varpi : (P, G) \rightarrow (D, g)$ is a Riemannian submersion. For $\Pi \subset T_{[p]\star} D$, its horizontal lift is $\tilde{\Pi} = (d\varpi_p|_{\mathcal{H}_p^\star})^{-1}(\Pi)$. O'Neill's formula [11] gives $\sec_g(\Pi) \geq \sec_G(\tilde{\Pi})$, so positivity on these lifted two-planes suffices.

Section 3 proves the curvature criterion for the southern construction. Write Ω for the curvature of ω and $D\Omega$ for its covariant derivative, using ∇^{g_B} and the induced adjoint-bundle connection. With g_B, ω, Q fixed, set $r_\varepsilon^2 = \varepsilon e^{\varepsilon\varphi}$. A sufficiently strong lower bound on $-\text{Hess } \varphi$, chosen in terms of $\Omega, D\Omega$ before ε , supplies positive mixed curvature. The Lie-bracket terms linear in Ω have no small fibre factor; on decomposable bivectors, they couple horizontal and vertical areas and are absorbed by the positive contributions from those areas. For a positively curved base and sufficiently small ε , the criterion gives $\sec_{G_\varepsilon} > 0$ on every two-plane.

Section 4 applies it with a round base disk of radius less than $\pi/2$, a multiple of a round height function, and a $\star$ -invariant connection that is a product connection in the boundary trivialization. Theorem 4.1 gives g_S on D_S .

For $P_N \cong D^7 \times S^3$, Section 5 uses the flat product connection $\omega = u^{-1}du$ and the metric

$$G_N = ds^2 + F(s)^2 h_{S^6} + r(s)^2 h_{S^3},$$

where s is radial distance and h_{S^j} is the unit round metric. The function r is constant near the center and matches the southern boundary radius and logarithmic derivative. The ϖ_N -horizontal space is a graph over the base tangent space, so its fibre components are bounded in terms of its base components. This estimate allows the base curvature to dominate the radial-fibre and angular-fibre terms on these planes. The angular profile F is fixed before the common ε ; Theorem 5.1 then gives $\sec_{G_N} > 0$ on the ϖ_N -horizontal two-planes and hence $\sec_{g_N} > 0$ on D_N .

Finally, let h_N, h_S be the induced boundary metrics and B_N, B_S the outward second fundamental forms of $(D_N, g_N), (D_S, g_S)$.

Section 6 compares them under ϕ_n . In the common boundary trivialization, the source boundary metrics agree and the fibre radii have the same derivative in a signed normal coordinate across the seam. The fibre contributions to the outward second fundamental forms therefore cancel, while the angular contributions have a positive definite sum. Proposition 6.1 proves

$$h_N = \phi_n^* h_S, \quad B_N + \phi_n^* B_S > 0,$$

which completes Theorem 1.3.

The Reiser–Wraith gluing theorem [14, Theorem A(i), $k = 1$] then yields a smooth positively curved metric on $D_N \cup_{\phi_n} D_S \cong \Sigma_n$. Appendix B includes a proof of the strict gluing statement used here.

1.3. Conventions and notation. Our curvature convention is

$$R(A, B)C = \nabla_A \nabla_B C - \nabla_B \nabla_A C - \nabla_{[A, B]} C.$$

For a decomposable bivector $\xi = A \wedge B$, put

$$\mathcal{K}(\xi) = \langle R(A, B)B, A \rangle.$$

Thus $\sec(\text{span}\{A, B\}) = \mathcal{K}(\xi)/|\xi|^2$, and the unit round sphere has curvature one. Whenever curvature is regarded as a bilinear form on Λ^2 , our convention is $\mathcal{R}(A \wedge B, C \wedge D) = -\langle R(A, B)C, D \rangle$. For a boundary,

$$(1.2) \quad B(Y, Z) = \langle \nabla_Y n_{\text{out}}, Z \rangle.$$

If ρ is inward distance and the metric is $d\rho^2 + h_\rho$, then $B = -\frac{1}{2}\partial_\rho h_\rho$ at the boundary.

We identify S^3 with the unit quaternions. Its Lie algebra $\text{Im } \mathbb{H}$ has the Euclidean inner product Q , so its bi-invariant metric has sectional curvature one and $[u, v] = 2u \times v$. The symbol h_{S^j} denotes the unit round metric on S^j . Frame indices i, j, k are horizontal and a, b, c are vertical, with $1 \leq a, b, c \leq 3$. Repeated frame indices in displayed contractions are summed over their full ranges unless a restricted sum is specified.

Exterior powers carry their induced Euclidean norms. In particular, for an adjoint-bundle-valued two-form we use

$$(1.3) \quad |\Omega|^2 = \sum_{i < j, a} (\Omega_{ij}^a)^2, \quad |D\Omega|^2 = \sum_{k, i < j, a} ((D_k \Omega)_{ij}^a)^2.$$

Consequently their full ordered component arrays have norms $\sqrt{2}|\Omega|$ and $\sqrt{2}|D\Omega|$. A mixed tensor $\beta \in H \otimes V$ has norm $|\beta|^2 = \sum_{i,a} \beta_{ia}^2$. Other tensor norms are the induced Hilbert–Schmidt norms; operator norms are marked explicitly. Angular and fibre components are taken in orthonormal frames for the scaled metric. The maps K_x and K_x^* introduced in Section 5 are measured in the unit-round angular metric and the Lie algebra inner product Q .

Code. A reproducible SageMath verification package accompanies this paper, providing exact symbolic checks of algebraic and curvature identities, coordinate and boundary compatibility formulas, and the arithmetic underlying parameter choices and error estimates, together with rigorous interval checks of scalar bounds. The source code, exact-arithmetic certificates, independent certificate checker, and a detailed verification report with reproduction instructions are freely available at <https://zliu-math.github.io/projects/PosSecSmoothS7>.

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2. THE MARKED QUATERNIONIC MODELS

We describe the principal bundles and star quotients used below, with fixed trivializations and quotient disk coordinates. We then compare their attaching maps with the two-disk models of Durán, Püttmann and Rigas [3]. These coordinates will be used in Sections 4–6.

For each $n \in \mathbb{Z}$, we specify a smooth principal right S^3 -bundle

$$\pi_n : P_n \longrightarrow S^7$$

and two commuting S^3 -actions on its ten-dimensional total space: the *principal right action* and a smooth free left action, called the *star action* and denoted by $\star$. The quotient by the principal right action is the base S^7 . We write

$$\Sigma_n := P_n / S_\star^3$$

for the orbit space of the star action and give explicit smooth coordinates identifying it as

$$\Sigma_n \cong D_N \cup_{\phi_n} D_S,$$

where D_N, D_S are smooth seven-disks and $\phi_n : \partial D_N \rightarrow \partial D_S$ is the attaching diffeomorphism computed below. The identification of this quotient with the class $n \bmod 28$ in Θ_7 is the conclusion of [3, Theorem 1]; we verify its application by comparing the attaching map in our coordinate realization with the one in their Section 4.

The resulting transition formula is used to choose the southern connection potentials with the required behavior near the boundary and the south pole in Section 4. The quotient disk coordinates give the boundary markings used in Section 5. Their change of coordinates specifies the attaching map under which Section 6 compares the boundary metrics and outward second fundamental forms.

2.1. Quaternionic coordinates and conjugation. Let $\mathbb{H}$ denote the algebra of quaternions, regarded as the real Euclidean space $\mathbb{R}^4$. For $v \in \mathbb{H}$, write $\bar{v}$ for its quaternionic conjugate and $|v| = (v\bar{v})^{1/2}$ for its norm. Set

$$\text{Im } \mathbb{H} = \{v \in \mathbb{H} : \bar{v} = -v\}, \quad S^3 = \{q \in \mathbb{H} : |q| = 1\}.$$

Thus $\text{Im } \mathbb{H}$ is a three-dimensional real vector space, and S^3 is a group under quaternionic multiplication, with $q^{-1} = \bar{q}$. The seven-dimensional real vector space

$$\mathcal{V} = \text{Im } \mathbb{H} \oplus \mathbb{H}$$

has elements $x = (p, w)$ with norm $|x|^2 = |p|^2 + |w|^2$. Its unit sphere is

$$S^6 = \{(p, w) \in \mathcal{V} : |p|^2 + |w|^2 = 1\}.$$

Here $p \in \text{Im } \mathbb{H}$ is purely imaginary, whereas w is an arbitrary quaternion.

For $q \in S^3$ and $x = (p, w) \in \mathcal{V}$, define simultaneous conjugation by

$$C_q(p, w) = (qpq^{-1}, qwq^{-1}).$$

Quaternionic conjugation by q preserves norms and the imaginary subspace, so C_q maps S^6 to itself.

The identities $C_1 = \text{Id}$ and $C_{q_1 q_2} = C_{q_1} \circ C_{q_2}$ make this a *left* S^3 -action on $\mathcal{V}$. We abbreviate $C_q(x)$ to qxq^{-1} ; products with a pair $x = (p, w)$ are always taken componentwise. We also let S^3 act on itself by $v \mapsto qvq^{-1}$.

A map $f : S^6 \rightarrow S^3$ is *equivariant* for these actions when

$$f(qxq^{-1}) = qf(x)q^{-1} \quad (q \in S^3, x \in S^6).$$

2.2. The equivariant Blakers–Massey map. For an imaginary quaternion $p \in \text{Im } \mathbb{H}$, the quaternionic exponential is

$$\exp(\pi p) = \cos(\pi|p|) + \frac{\sin(\pi|p|)}{|p|} p,$$

where the scalar factor $\sin(\pi|p|)/|p|$ is assigned its limiting value π at $p = 0$; in particular, $\exp(0) = 1$. The Blakers–Massey map $b : S^6 \rightarrow S^3$ used in [2, 3] is given, for $w \neq 0$, by

$$(2.1) \quad b(p, w) = \frac{w}{|w|} \exp(\pi p) \frac{\bar{w}}{|w|}.$$

The factors $w/|w|$ and $\exp(\pi p)$ are unit quaternions, and $\bar{w}/|w| = (w/|w|)^{-1}$, so this formula takes values in S^3 .

To define b at $w = 0$ and verify smoothness, put $s = |p|^2$. On S^6 we have $|w|^2 = 1 - s$, and (2.1) becomes

$$b(p, w) = \cos(\pi\sqrt{s}) + \psi(s)wp\bar{w}, \quad \text{with} \quad \psi(s) = \frac{\sin(\pi\sqrt{s})}{\sqrt{s}(1-s)} \quad (0 < s < 1).$$

Here ψ is a real-valued auxiliary function. It extends smoothly to both endpoints, with

$$\psi(0) = \pi, \quad \psi(1) = \frac{\pi}{2}.$$

Indeed, near $s = 0$,

$$\frac{\sin(\pi\sqrt{s})}{\sqrt{s}} = \sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k+1}}{(2k+1)!} s^k,$$

while near $s = 1$ its numerator has the expansion

$$\sin(\pi\sqrt{s}) = \frac{\pi}{2}(1-s) + O((1-s)^2).$$

The zero at $s = 1$ therefore removes the factor $1 - s$ from the denominator. The function $\cos(\pi\sqrt{s})$ is also smooth at $s = 0$, as its power series contains only integer powers of s . Consequently the displayed expression for the Blakers–Massey map b is smooth on all of S^6 . It gives $b(p, 0) = -1$ when $|p| = 1$, and $b(0, w) = 1$ when $|w| = 1$. Its values remain in S^3 by continuity.

The identities

$$\exp(\pi qpq^{-1}) = q \exp(\pi p) q^{-1}, \quad \overline{qwq^{-1}} = q\bar{w}q^{-1}$$

show that b is equivariant when $w \neq 0$; continuity gives the same identity at $w = 0$.

For each $n \in \mathbb{Z}$, define the n -th pointwise power of the Blakers–Massey map

$$g_n : S^6 \longrightarrow S^3, \quad g_n(x) = b(x)^n.$$

The powers here are taken pointwise in the group S^3 : $g_0 \equiv 1$, and $g_{-m}(x) = (b(x)^{-1})^m$ for $m > 0$. Since multiplication and inversion on S^3 are smooth, each g_n is smooth. Equivariance of b gives

$$(2.2) \quad g_n(qxq^{-1}) = qg_n(x)q^{-1}, \quad x \in S^6, \quad q \in S^3.$$

2.3. The principal bundle P_n . Use the model

$$S^7 = \{(x_0, z) \in \mathbb{R} \oplus \mathcal{V} : x_0^2 + |z|^2 = 1\},$$

with north and south poles $o_N = (1, 0)$ and $o_S = (-1, 0)$. The polar parameter is $t = \arccos x_0 \in [0, \pi]$. Away from the poles, write a base point uniquely as

$$(x_0, z) = (\cos t, \sin t x), \quad 0 < t < \pi, \quad x \in S^6.$$

Let $\mathcal{U}_N = S^7 \setminus \{o_S\}$ and $\mathcal{U}_S = S^7 \setminus \{o_N\}$. These are polar coordinate neighborhoods: their disk coordinates are tx and $(\pi - t)x$, respectively, with the corresponding pole represented by the zero vector.

Take the product bundles $\mathcal{U}_N \times S^3$ and $\mathcal{U}_S \times S^3$. Denote their fibre coordinates by $u_N, u_S \in S^3$. For a common base point (t, x) in the overlap, identify the two fibre coordinates by

$$(2.3) \quad u_S = g_n(x)u_N.$$

Thus (t, x, u_N) and $(t, x, g_n(x)u_N)$ are representatives of the same point of the glued total space P_n . Appendix A gives the corresponding quotient-space construction and its smooth principal-bundle structure.

The projection $\pi_n : P_n \rightarrow S^7$ sends either representative to $(\cos t, \sin t x)$. The transition map is smooth, with inverse $u_N = g_n(x)^{-1}u_S$, so these product charts define a smooth ten-dimensional bundle with fibre S^3 .

2.4. Two actions of S^3 on P_n . The *principal right action* is right multiplication in each fibre. In either chart it is

$$(t, x, u) \cdot h = (t, x, uh), \quad h \in S^3.$$

It respects (2.3) because $g_n(x)(u_N h) = (g_n(x)u_N)h$. It fixes the base point and is free and transitive on each fibre: two points of a fibre differ by right multiplication by a unique h . This is the principal right S^3 -bundle structure on P_n .

The second action, called the *star action*,

$$\star : S^3 \times P_n \longrightarrow P_n, \quad (q, v) \longmapsto q \star v,$$

is given in either polar trivialization, away from the poles, by

$$(2.4) \quad q \star (t, x, u) = (t, qxq^{-1}, qu), \quad q \in S^3.$$

This is a left action: $1 \star v = v$ and $q_1 \star (q_2 \star v) = (q_1 q_2) \star v$. In a polar disk coordinate $\zeta = \rho x$, the same formula reads

$$q \star (\zeta, u) = (q\zeta q^{-1}, qu),$$

which is smooth also at $\zeta = 0$. To check compatibility with the transition, use (2.2):

$$g_n(qxq^{-1})(qu_N) = qg_n(x)u_N = qu_S.$$

Hence the local formulas define an action on all of P_n . It is free, since $qu = u$ implies $q = 1$, including over the poles. It commutes with the principal right action, since

$$q \star ((\zeta, u) \cdot h) = (q\zeta q^{-1}, quh) = (q \star (\zeta, u)) \cdot h.$$

Unlike the right action, this second action generally changes the base point: it covers $(x_0, z) \mapsto (x_0, qzq^{-1})$ on S^7 .

2.5. Quotient disks and boundary markings. For $v, v' \in P_n$, define

$$v \sim_\star v' \iff v' = q \star v \text{ for some } q \in S^3.$$

We write

$$\Sigma_n := P_n / \sim_\star = P_n / S_\star^3, \quad \varpi_n : P_n \longrightarrow \Sigma_n, \quad v \longmapsto [v]_\star = \{q \star v : q \in S^3\}.$$

Thus $S_\star^3$ specifies the action used to form the orbit space, rather than a different group. The quotient by the principal right action is the base S^7 ; the quotient by $\star$ is Σ_n . We now give explicit smooth coordinates for the latter quotient.

Choose $0 < a < \pi$, and let $U_N = \{0 \leq t \leq a\}$ and $U_S = \{a \leq t \leq \pi\}$ be the two closed polar disks in S^7 . The cut value a will be chosen in Section 4. For $\alpha \in \{N, S\}$, set

$$P_\alpha = \pi_n^{-1}(U_\alpha), \quad D_\alpha = P_\alpha / S_\star^3.$$

Each P_α is invariant under $\star$. Its disk coordinate is $\zeta_\alpha = \rho_\alpha x$, where $\rho_N = t$ and $\rho_S = \pi - t$. Put $R_N = a$, $R_S = \pi - a$, and write $\mathbb{D}(R) = \{z \in \mathcal{V} : |z| \leq R\}$. In the product trivialization of P_α , define

$$F_\alpha(\zeta_\alpha, u_\alpha) = u_\alpha^{-1} \zeta_\alpha u_\alpha \in \mathbb{D}(R_\alpha).$$

This map is constant on star orbits, since

$$(qu_\alpha)^{-1}(q\zeta_\alpha q^{-1})(qu_\alpha) = u_\alpha^{-1} \zeta_\alpha u_\alpha.$$

Every orbit has a unique representative of the form $(Y, 1)$: acting on (ζ, u) by $q = u^{-1}$ gives $(u^{-1}\zeta u, 1)$, and an action preserving the second coordinate 1 must have $q = 1$. Therefore F_α induces a diffeomorphism

$$D_\alpha \longrightarrow \mathbb{D}(R_\alpha), \quad [(\zeta_\alpha, u_\alpha)]_\star \longmapsto Y_\alpha = u_\alpha^{-1} \zeta_\alpha u_\alpha,$$

with inverse $Y \mapsto [(Y, 1)]_\star$. These formulas are smooth at the disk center as well as away from it. They identify each D_α with a smooth seven-disk; the transition between the two quotient coordinate systems is computed below.

The boundary of each quotient disk is parameterized by the unit sphere S^6 using

$$y_\alpha = R_\alpha^{-1} Y_\alpha = u_\alpha^{-1} x u_\alpha \in S^6.$$

These are the boundary markings. Let $\phi_n : \partial D_N \rightarrow \partial D_S$ be the identification induced by the bundle transition (2.3); then $\Sigma_n \cong D_N \cup_{\phi_n} D_S$. Let $\mathbb{D}_N^7$ and $\mathbb{D}_S^7$ be labelled copies of $\mathbb{D}(1) \subset \mathcal{V}$. Write the normalized quotient charts as

$$\Psi_\alpha : D_\alpha \longrightarrow \mathbb{D}_\alpha^7, \quad \Psi_\alpha([(\zeta_\alpha, u_\alpha)]_\star) = R_\alpha^{-1} u_\alpha^{-1} \zeta_\alpha u_\alpha, \quad \alpha \in \{N, S\},$$

and let $m_\alpha := \Psi_\alpha|_{\partial D_\alpha}$ denote their boundary restrictions. These are the markings $y_\alpha = u_\alpha^{-1} x u_\alpha$ defined above. The following lemma computes the attaching map in these coordinates. This is the coordinate comparison needed to apply [3, Theorem 1 and its proof in Section 4].

Lemma 2.1 (The attaching map). *The map*

$$\sigma : S^6 \longrightarrow S^6, \quad \sigma(x) = b(x)^{-1} x b(x),$$

is a diffeomorphism, and the boundary markings satisfy

$$m_S \circ \phi_n \circ m_N^{-1} = \sigma^n.$$

The quotient charts Ψ_N, Ψ_S induce a smooth diffeomorphism

$$J_n : \Sigma_n \longrightarrow M_n := \mathbb{D}_N^7 \cup_{\sigma^n} \mathbb{D}_S^7.$$

Proof. For representatives of the same point over the common boundary, (2.3) gives $u_S = g_n(x)u_N$. By equivariance, $g_n(u_N^{-1} x u_N) = u_N^{-1} g_n(x) u_N$. Thus

$$\begin{aligned} y_S &= u_N^{-1} g_n(x)^{-1} x g_n(x) u_N \\ &= g_n(y_N)^{-1} y_N g_n(y_N). \end{aligned}$$

Set $\hat{\sigma}(x) = b(x) x b(x)^{-1}$. Equivariance of b gives

$$b(\sigma(x)) = b(x), \quad b(\hat{\sigma}(x)) = b(x).$$

Both maps are smooth, and substitution yields $\hat{\sigma} \circ \sigma = \sigma \circ \hat{\sigma} = \text{Id}_{S^6}$. Hence σ is a diffeomorphism. Iterating σ and its inverse, respectively, gives

$$\sigma^n(x) = b(x)^{-n} x b(x)^n = g_n(x)^{-1} x g_n(x) \quad (n \in \mathbb{Z}).$$

Substitution in the boundary calculation gives

$$(2.5) \quad y_S = \sigma^n(y_N),$$

which is precisely $m_S \circ \phi_n \circ m_N^{-1} = \sigma^n$.

The charts Ψ_N, Ψ_S therefore descend to the stated bijection J_n . Their smooth inverses on the individual disks, $y \mapsto [(R_\alpha y, 1)]_\star$, agree on the identified boundaries: for $y \in S^6$,

$$[(R_N y, 1)_N]_\star = [(R_S y, g_n(y))_S]_\star = [(R_S \sigma^n(y), 1)_S]_\star,$$

where the subscripts indicate the product trivializations. On the open polar overlap, the unnormalized quotient coordinates satisfy

$$Y_S = (\pi - |Y_N|) \sigma^n \left(\frac{Y_N}{|Y_N|} \right), \quad 0 < |Y_N| < \pi.$$

To check smoothness across the seam at $t = a$, let ρ be the radial coordinate in either unit disk. Use $R_N(\rho - 1)$ on the northern collar and $R_S(1 - \rho)$ on the southern collar as the signed collar parameter; both pull back to $t = a$. Use y_N on the northern side and $\sigma^{-n}(y_S)$ on the southern side as the angular coordinate. In these collars, J_n and its inverse preserve both coordinates, so they are smooth across the seam. $\square$

The proof of [3, Theorem 1, Section 4] presents their sphere Σ_n^{DPR} as two polar disks attached by the n -th iterate of

$$(p, w) \mapsto \overline{b(p, w)}(p, w) b(p, w).$$

This is the same σ , since $\overline{b(p, w)} = b(p, w)^{-1}$. To compare the cut $t = a$ with their cut at $t = \pi/2$, use a smooth increasing diffeomorphism of $[0, \pi]$ fixing its endpoints, carrying a to $\pi/2$, and linear near the endpoints. This radial change preserves the angular attaching map and is smooth at the disk centers and across the seam. It identifies M_n with Σ_n^{DPR} . Give M_n the orientation induced by this comparison and orient Σ_n by pullback through J_n . Applying [3, Theorem 1] then gives

$$[\Sigma_n] = [\Sigma_n^{\text{DPR}}] = n[\Sigma_{\text{GM}}] \quad \text{in } \Theta_7 \cong \mathbb{Z}/28\mathbb{Z},$$

with the orientation of Σ_{GM} chosen to represent 1. Here Θ_7 denotes the group of oriented diffeomorphism classes of homotopy seven-spheres. Since its group operation is oriented connected sum,

$$\Sigma_2 \cong_{\text{or}} \Sigma_{\text{GM}} \# \Sigma_{\text{GM}},$$

where $\cong_{\text{or}}$ denotes an orientation-preserving diffeomorphism.

If the star action is forgotten, the principal bundle P_n is classified by the homotopy class $[g_n] = n[b]$ of its transition map in $\pi_6(S^3) \cong \mathbb{Z}/12\mathbb{Z}$. A principal-bundle isomorphism respects the right action but need not respect the star action. Thus isomorphic principal bundles can have different star quotients; in particular, some exotic quotients arise from trivial principal bundles. This distinction is described in [3, Corollary 2.2 and Remark 2.5].

Remark 2.2 (Boundary coordinates). On a collar of the southern boundary, we use the north product trivialization. Its quotient angular coordinate is y_N , whereas the extending south trivialization gives y_S . Equation (2.5) relates these angular coordinates throughout the overlap. Hence the boundary identification is the identity when both sides are expressed in the common north trivialization, and is represented by σ^n in the extending north and south

disk markings. Section 6 compares the boundary metrics and second fundamental forms under this map.

3. POSITIVE CONNECTION METRICS OVER A DISK

The southern construction requires a connection metric with positive sectional curvature. The next proposition gives a sufficient condition in terms of the Hessian of the fibre radius and the curvature of a fixed connection.

Let $\pi : P \rightarrow (B, g_B)$ be a principal right S^3 -bundle, with a fixed smooth connection ω and curvature Ω . For a positive function r on B , consider

$$(3.1) \quad G = \pi^* g_B + r^2 Q(\omega, \omega).$$

In this section, H and V denote the horizontal and vertical distributions of the principal projection π , respectively.

Proposition 3.1 (Positive connection metrics). *Suppose that B is compact, possibly with boundary, and $\sec_{g_B} \geq \kappa > 0$. Put*

$$M_0 = \sup_B |\Omega|, \quad M_1 = \sup_B |D\Omega|,$$

where D uses the base Levi-Civita connection and the induced adjoint-bundle connection, and the norms are (1.3). Set $C = 4$. If

$$(3.2) \quad \Lambda \geq 16CM_0^2 + 64C^2\kappa^{-1}(M_1 + 1)^2 + 4$$

and a fixed smooth function φ satisfies

$$-\text{Hess}_B \varphi \geq \Lambda g_B,$$

then the metrics

$$(3.3) \quad G_\varepsilon = \pi^* g_B + r_\varepsilon^2 Q(\omega, \omega), \quad r_\varepsilon^2 = \varepsilon e^{\varepsilon \varphi},$$

have strictly positive sectional curvature for all sufficiently small $\varepsilon > 0$.

We first compute the curvature of (3.1) and estimate it on decomposable bivectors. The concavity condition (3.2) is imposed before choosing ε : after Young's inequality, the terms involving $D\Omega$ contribute a loss of order $\varepsilon(M_1 + 1)^2$, the same order in ε as the positive Hessian term. On the southern disk, φ will be a multiple of a round height function.

Remark 3.2 (Mixed curvature and the fibre radius). Variation of the fibre radius can give positive mixed curvature even for a flat connection. For example,

$$d\rho^2 + \sin^2 \rho h_{S^6} + \cos^2 \rho h_{S^3}, \quad 0 \leq \rho \leq R < \pi/2,$$

is the round metric of curvature one on a tube around a great $S^3 \subset S^{10}$. Viewed as a connection metric over a seven-disk, it has a flat connection and mixed curvature $-r^{-1} \text{Hess } r = g_B$.

3.1. The curvature components. Choose a base orthonormal frame normal at a given point, and let X_i be its principal-horizontal lifts. For a fixed Q -orthonormal basis e_a of $\text{Im } \mathbb{H}$, write

$$E_a = r^{-1} e_a^\#, \quad \vartheta = d \log r, \quad [e_a, e_b] = c_{ab}^c e_c.$$

The fields $e_a^\#$ are the fundamental fields of the principal right action, with $e_a^\#(p) = \frac{d}{dt} \Big|_{t=0} p \exp(te_a)$ for $p \in P$. At the chosen point, the relevant bracket identities are

$$(3.4) \quad [X_i, X_j]^V = -r\Omega_{ij}^a E_a, \quad [X_i, E_a] = -\vartheta_i E_a, \quad [E_a, E_b] = r^{-1} c_{ab}^c E_c.$$

Since the curvature form is adjoint-equivariant,

$$(3.5) \quad E_a(\Omega_{ij}^b) = -r^{-1}c_{ac}^b\Omega_{ij}^c.$$

Koszul's formula gives

$$(3.6) \quad \nabla_{X_i}X_j = (\nabla_i^B X_j)^H - \frac{r}{2}\Omega_{ij}^a E_a, \quad \nabla_{X_i}E_a = \frac{r}{2}\Omega_{ij}^a X_j,$$

$$(3.7) \quad \nabla_{E_a}X_i = \frac{r}{2}\Omega_{ij}^a X_j + \vartheta_i E_a, \quad \nabla_{E_a}E_b = \frac{1}{2r}c_{ab}^c E_c - \delta_{ab}\vartheta_i X_i.$$

We differentiate these identities in the chosen frame and then evaluate at the base point.

Set $N = -r^{-1}\text{Hess}_B r$, viewed as a symmetric endomorphism using g_B . We write $\Omega(X_i, X_j) = \Omega_{ij}^a e_a$ in the fixed Q -orthonormal Lie algebra basis. The curvature components involving both distributions are

$$(3.8) \quad \langle R(X_i, E_a)E_b, X_j \rangle = N_{ij}\delta_{ab} + \frac{r^2}{4} \sum_k \Omega_{ik}^b \Omega_{jk}^a - \frac{1}{4} \langle [e_a, e_b], \Omega_{ij} \rangle,$$

$$(3.9) \quad \langle R(X_i, X_j)E_a, E_b \rangle = \frac{r^2}{4} \sum_k (\Omega_{ik}^a \Omega_{jk}^b - \Omega_{jk}^a \Omega_{ik}^b) + \frac{1}{2} \langle [e_a, e_b], \Omega_{ij} \rangle,$$

$$(3.10) \quad \langle R(X_i, E_a)E_b, E_c \rangle = \frac{r}{2} \sum_k \vartheta_k (\delta_{ab}\Omega_{ik}^c - \delta_{ac}\Omega_{ik}^b),$$

$$(3.11) \quad \begin{aligned} \langle R(X_i, X_j)X_k, E_a \rangle &= -\frac{r}{2} \left((D_i \Omega)_{jk}^a - (D_j \Omega)_{ik}^a \right) \\ &\quad - \frac{r}{2} (\vartheta_i \Omega_{jk}^a - \vartheta_j \Omega_{ik}^a) + r \vartheta_k \Omega_{ij}^a. \end{aligned}$$

For (3.8), the relevant horizontal components at the chosen point are

$$\begin{aligned} \langle \nabla_{X_i} \nabla_{E_a} E_b, X_j \rangle &= \frac{1}{4} c_{ab}^c \Omega_{ij}^c - \delta_{ab} \partial_i \vartheta_j, \\ \langle \nabla_{E_a} \nabla_{X_i} E_b, X_j \rangle &= -\frac{1}{2} c_{ac}^b \Omega_{ij}^c + \frac{r^2}{4} \sum_k \Omega_{ik}^b \Omega_{kj}^a, \\ \langle \nabla_{[X_i, E_a]} E_b, X_j \rangle &= \delta_{ab} \vartheta_i \vartheta_j. \end{aligned}$$

Use $c_{ac}^b = -c_{ab}^c$, $\Omega_{kj} = -\Omega_{jk}$, and $N_{ij} = -\partial_i \vartheta_j - \vartheta_i \vartheta_j$ to obtain (3.8). If its left side is denoted by M_{ijab} , the first Bianchi identity gives

$$\langle R(X_i, X_j)E_a, E_b \rangle = M_{jiab} - M_{ijab},$$

which proves (3.9).

For (3.10), differentiating the vertical term $c_{ab}^c/(2r)$ in $\nabla_{E_a} E_b$ contributes $-\vartheta_i c_{ab}^c/(2r)$. The bracket term $-\nabla_{[X_i, E_a]} E_b$ contributes its opposite. The remaining vertical components are

$$\frac{r}{2} \delta_{ab} \sum_k \vartheta_k \Omega_{ik}^c \quad \text{and} \quad -\frac{r}{2} \delta_{ac} \sum_k \vartheta_k \Omega_{ik}^b.$$

Their sum gives (3.10). For (3.11), the vertical parts of $\nabla_{X_i} \nabla_{X_j} X_k - \nabla_{X_j} \nabla_{X_i} X_k$ are the first two terms of (3.11). The vertical part of $-\nabla_{[X_i, X_j]} X_k$ is $r \vartheta_k \Omega_{ij}^a E_a$, giving its last term.

The purely horizontal sectional numerator is

$$(3.12) \quad \mathcal{K}_G(X \wedge Y) = \mathcal{K}_B(X \wedge Y) - \frac{3}{4} r^2 |\Omega(X, Y)|^2.$$

This is the horizontal submersion formula [11], since the vertical bracket is $-r\Omega(X, Y)^a E_a$. The fibres are round with radius r ; their second fundamental form is $-G|_V \otimes \nabla \log r$, by (3.7).

Their Gauss equation therefore gives the vertical curvature tensor of constant-curvature type with coefficient

$$(3.13) \quad r^{-2} - |\vartheta|^2.$$

We now combine these components to estimate the sectional curvature of an arbitrary two-plane.

3.2. A lower bound for arbitrary planes. Write a decomposable bivector as

$$\xi = (X + U) \wedge (Y + V) = \alpha + \beta + \gamma,$$

where

$$\alpha = X \wedge Y, \quad \beta = X \otimes V - Y \otimes U, \quad \gamma = U \wedge V.$$

Here X, Y are principal-horizontal and U, V are principal-vertical. We identify $H \wedge V$ with $H \otimes V$ by $X \wedge U \mapsto X \otimes U$. All components are taken in the normalized frames above. In expressions involving $[U, V]$, the vertical coefficient vectors are identified with $\text{Im } \mathbb{H}$ by $E_a \leftrightarrow e_a$. Set

$$h = |\alpha|, \quad m = |\beta|, \quad k = |\gamma|.$$

Then $|\xi|^2 = h^2 + m^2 + k^2$.

The terms linear in Ω are estimated using the skew-symmetry of the Lie bracket. Since ξ is decomposable, its components satisfy

$$(3.14) \quad \beta_{ia}\beta_{jb} - \beta_{ib}\beta_{ja} = \alpha_{ij}\gamma_{ab}.$$

Put $\mathcal{J} = \langle \Omega(X, Y), [U, V] \rangle$. The ordered-index contraction is

$$\sum_{i,j,a,b,c} c_{ab}^c \Omega_{ij}^c \beta_{ia} \beta_{jb} = \frac{1}{2} \sum_{i,j,a,b,c} c_{ab}^c \Omega_{ij}^c \alpha_{ij} \gamma_{ab} = 2\mathcal{J}.$$

The first equality uses skew-symmetry in a, b and (3.14); the second uses both skew index pairs. Consequently the bracket part of the $\beta\beta$ term in (3.8) is $-\mathcal{J}/2$. For the off-diagonal block, our curvature-form convention and (3.9) give

$$2 \sum_{\substack{i < j \\ a < b}} \alpha_{ij} \gamma_{ab} \left(-\frac{1}{2} \sum_c c_{ab}^c \Omega_{ij}^c \right) = -\mathcal{J}.$$

Here the leading factor two is the cross-block factor, and the minus sign comes from $\mathcal{R}(A \wedge B, C \wedge D) = -\langle R(A, B)C, D \rangle$. The sum of the two contributions is therefore

$$(3.15) \quad -\frac{3}{2} \langle \Omega(X, Y), [U, V] \rangle.$$

Since $|[U, V]| = 2|U \wedge V|$ and $|\Omega(X, Y)| \leq M_0|X \wedge Y|$, the absolute value of (3.15) is at most $3M_0hk$.

The N -term in (3.8) contributes

$$\sum_{i,j,a} N_{ij} \beta_{ia} \beta_{ja} \geq \lambda_{\min}(N) m^2.$$

For the remaining blocks, we use Cauchy-Schwarz with the norms in (1.3). The purely horizontal term is at least $(\kappa - \frac{3}{4}r^2 M_0^2)h^2$; only the sectional lower bound of the base is used, since $\alpha = X \wedge Y$ is decomposable. The purely vertical contribution is exactly $(r^{-2} - |\vartheta|^2)k^2$.

For the quadratic part of (3.8), Cauchy–Schwarz gives

$$\left(\sum_{i,j,a,b} \left(\sum_k \Omega_{ik}^b \Omega_{jk}^a \right)^2 \right)^{1/2} \leq \sum_{i,k,a} (\Omega_{ik}^a)^2 = 2|\Omega|^2.$$

Its contribution therefore has absolute value at most $\frac{1}{2}r^2M_0^2m^2$. The full ordered-array norm of the quadratic part of (3.9) is at most $r^2M_0^2$. Restricting both skew pairs to $i < j$ and $a < b$ divides that norm by two. The factor two from the off-diagonal block in the curvature quadratic form then gives the bound $r^2M_0^2hk$.

The full ordered-array norm of (3.11) is bounded by

$$\sqrt{2}rM_1 + 2\sqrt{2}r|\vartheta|M_0.$$

Restricting its first skew pair to $i < j$ divides this bound by $\sqrt{2}$. Including the cross-block factor two, its contribution is bounded by $2r(M_1 + 2|\vartheta|M_0)hm$. For (3.10), put $w_i^a = \sum_k \vartheta_k \Omega_{ik}^a$. Then

$$\sum_{i,a} (w_i^a)^2 \leq 2|\vartheta|^2|\Omega|^2, \quad \sum_{a,b,c} (\delta_{ab}w_i^c - \delta_{ac}w_i^b)^2 = 4|w_i|^2.$$

After restricting $b < c$ and including the cross-block factor two, its contribution is at most $2r|\vartheta|M_0km$. Combining the three diagonal and three off-diagonal blocks in

$$\Lambda^2(H \oplus V) = \Lambda^2H \oplus (H \otimes V) \oplus \Lambda^2V$$

with the bracket bound $3M_0hk$ gives

$$\begin{aligned} \mathcal{K}_G(\xi) &\geq (\kappa - \tfrac{3}{4}r^2M_0^2)h^2 + (\lambda_{\min}N - \tfrac{1}{2}r^2M_0^2)m^2 + (r^{-2} - |\vartheta|^2)k^2 \\ &\quad - 2r(M_1 + 2|\vartheta|M_0)hm - (3M_0 + r^2M_0^2)hk - 2r|\vartheta|M_0km. \end{aligned} \quad (3.16)$$

In particular, setting $C = 4$ yields the convenient common-constant estimate

$$\begin{aligned} \mathcal{K}_G(\xi) &\geq (\kappa - Cr^2M_0^2)h^2 + (r^{-2} - |\vartheta|^2)k^2 + (\lambda_{\min}N - Cr^2M_0^2)m^2 \\ &\quad - Cr(M_1 + |\vartheta|M_0)hm - C(M_0 + r^2M_0^2)hk - Cr|\vartheta|M_0km. \end{aligned} \quad (3.17)$$

The connection enters this estimate through M_0 and M_1 .

Remark 3.3 (Decomposability). Identity (3.14) converts the terms linear in Ω into a coupling of the horizontal and vertical areas. This is why (3.17) contains M_0hk . The horizontal term is estimated using the decomposable bivector $X \wedge Y$, so a sectional-curvature lower bound on the base suffices throughout.

3.3. Choice of the fibre scale.

Proof of Proposition 3.1. Let $D_0 = \sup_B |d\varphi|$. For $r^2 = \varepsilon e^{\varepsilon\varphi}$,

$$\vartheta = \frac{\varepsilon}{2}d\varphi, \quad N = -\frac{\varepsilon}{2}\text{Hess } \varphi - \frac{\varepsilon^2}{4}d\varphi \otimes d\varphi.$$

Choose $\varepsilon > 0$ so small that

$$|\varepsilon\varphi| \leq \log 2, \quad 2\varepsilon M_0 \leq 1, \quad \varepsilon D_0 M_0 \leq 1, \quad (3.18)$$

$$8C\varepsilon M_0^2 \leq \kappa, \quad 64C^2\varepsilon M_0^2 \leq \kappa, \quad (3.19)$$

$$\varepsilon D_0^2 \leq \Lambda/4, \quad 2\varepsilon^3 D_0^2 \leq 1, \quad 32C^2\varepsilon^3 D_0^2 M_0^2 \leq \Lambda. \quad (3.20)$$

All quantities except ε have already been fixed. Thus these finitely many upper bounds admit a common positive solution.

We have $\varepsilon/2 \leq r^2 \leq 2\varepsilon$ and $|\vartheta| \leq \varepsilon D_0/2$. The first two coefficients in (3.17) are consequently at least $3\kappa/4$ and $3/(8\varepsilon)$. Young's inequality gives

$$\begin{aligned} Cr(M_1 + |\vartheta|M_0)hm &\leq \frac{\kappa}{8}h^2 + 8C^2\kappa^{-1}\varepsilon(M_1 + 1)^2m^2, \\ C(M_0 + r^2M_0^2)hk &\leq \frac{\kappa}{8}h^2 + 8C^2\kappa^{-1}M_0^2k^2 \leq \frac{\kappa}{8}h^2 + \frac{1}{8\varepsilon}k^2, \\ Cr|\vartheta|M_0km &\leq \frac{1}{8\varepsilon}k^2 + 2C^2\varepsilon^4D_0^2M_0^2m^2. \end{aligned}$$

The mixed coefficient starts at $\varepsilon\Lambda/2$. The loss from $d\varphi \otimes d\varphi$ is at most $\varepsilon\Lambda/16$; the loss $Cr^2M_0^2$ is at most $\varepsilon\Lambda/8$. The two Young losses in m^2 are at most $\varepsilon\Lambda/8$ and $\varepsilon\Lambda/16$, respectively. Using (3.2) and (3.20), we obtain

$$(3.21) \quad \mathcal{K}_{G_\varepsilon}(\xi) \geq \frac{\kappa}{2}h^2 + \frac{\varepsilon\Lambda}{8}m^2 + \frac{1}{8\varepsilon}k^2.$$

For a unit decomposable bivector, $h^2 + m^2 + k^2 = 1$. Hence

$$(3.22) \quad \sec_{G_\varepsilon} \geq c_S := \min\{\kappa/2, \varepsilon\Lambda/8, 1/(8\varepsilon)\} > 0.$$

For each fixed admissible $\varepsilon > 0$, the right-hand side is positive and independent of the point and two-plane. $\square$

4. THE SOUTHERN FILLING

The southern filling determines the boundary radii and fibre slope that the northern filling will match. We state the construction on the marked southern bundle of Section 2. Fix $n \in \mathbb{Z}$ and choose

$$\frac{\pi}{2} < a < b_0 < \pi.$$

The southern base is the round disk

$$U_S = \{a \leq t \leq \pi\}, \quad g_B = dt^2 + \sin^2 t h_{S^6}.$$

Its radius, measured from the south pole, is $\pi - a < \pi/2$. Choose a fixed smooth function $\chi(t) \in [0, 1]$ which is zero near a and one for $t \geq b_0$.

Theorem 4.1 (Southern filling). *Let ω be the connection specified by (4.4)–(4.5), and let M_0, M_1 be its curvature bounds as in Proposition 3.1. Choose Λ satisfying (3.2) with $\kappa = 1$, and fix $A_0 > 0$ such that*

$$A_0 |\cos a| \geq \Lambda.$$

There is $\varepsilon_S > 0$, depending only on these fixed data, such that for every $0 < \varepsilon < \varepsilon_S$ the metric

$$(4.1) \quad G_S = dt^2 + \sin^2 t h_{S^6} + r_S(t)^2 Q(\omega, \omega), \quad r_S(t)^2 = \varepsilon e^{-\varepsilon A_0 \cos t}$$

extends smoothly over the entire southern bundle and has strictly positive sectional curvature. The star action is free and isometric, and its quotient (D_S, g_S) is a smooth marked seven-disk. Both G_S and g_S satisfy (3.22) with $\kappa = 1$, including at the center and boundary.

Set

$$(4.2) \quad F_a = \sin a, \quad r_a^2 = \varepsilon e^{\varepsilon A_0 |\cos a|}, \quad q_s = \frac{r'_S(a)}{r_a} = \frac{\varepsilon A_0 F_a}{2}, \quad \mu_S = \cot a < 0.$$

In the north boundary gauge, $\omega = u^{-1}du$ on a boundary collar, and the induced source boundary metric is

$$F_a^2 h_{S^6} + r_a^2 h_{S^3}.$$

For a source boundary tangent vector $Z = X + U$, expressed in orthonormal angular and fibre components, the outward second fundamental form $\mathcal{B}_S$ is

$$(4.3) \quad \mathcal{B}_S(Z, Z) = -\mu_S |X|^2 - q_s |U|^2.$$

We construct the connection first, apply the curvature estimate, and then compute the boundary data in the north gauge.

4.1. The connection. In the north boundary gauge and the extending south gauge, respectively, define connection potentials

$$(4.4) \quad A_N = \chi g_n^{-1} dg_n, \quad A_S = (\chi - 1) dg_n g_n^{-1}.$$

They satisfy

$$A_N = g_n^{-1} A_S g_n + g_n^{-1} dg_n,$$

which is precisely the transformation rule associated with (2.3). The full principal connection form in either gauge is

$$(4.5) \quad \omega = u^{-1} A u + u^{-1} du.$$

Since $A_S = 0$ near the south pole, this defines a smooth connection on the entire southern bundle.

Put $\theta = g_n^{-1} dg_n$ and write $(\theta \wedge \theta)(X, Y) = [\theta(X), \theta(Y)]$. The curvature in the north gauge is

$$\Omega_N = d\chi \wedge \theta + \chi(\chi - 1)\theta \wedge \theta.$$

The connection is smooth on the compact base U_S , so M_0, M_1 are finite. These constants depend on n, a, b_0, χ and the fixed round base metric.

The connection is invariant under the star action. In fact, (2.2) gives $A \mapsto q A q^{-1}$ under conjugation of the base, while $u \mapsto q u$. For constant q ,

$$(qu)^{-1} (q A q^{-1}) (qu) + (qu)^{-1} d(qu) = u^{-1} A u + u^{-1} du.$$

The round base and radial functions are also invariant.

4.2. The metric and its quotient. The function $\varphi(t) = -A_0 \cos t$ satisfies

$$-\text{Hess}_B \varphi = -A_0 \cos t g_B \geq A_0 |\cos a| g_B.$$

Choose $\varepsilon_S > 0$ so that (3.18)–(3.20) hold with $\kappa = 1$ for every $0 < \varepsilon < \varepsilon_S$. Proposition 3.1 then gives positive sectional curvature for G_S , with the lower bound c_S in (3.22).

To express the metric at the south pole, set $\rho = \pi - t$. The radius becomes

$$\hat{r}_S(\rho) := r_S(\pi - \rho) = \sqrt{\varepsilon} e^{\varepsilon A_0 \cos \rho / 2},$$

a smooth even function. Write $r_{\text{pole}} := \hat{r}_S(0) = \sqrt{\varepsilon} e^{\varepsilon A_0 / 2}$. The connection potential vanishes in the extending south gauge there. At the pole, $\Omega = 0$, $\vartheta = d \log r_S = 0$, and $N = (\varepsilon A_0 / 2) g_B$; consequently the sectional numerator has the form

$$h^2 + \frac{\varepsilon A_0}{2} m^2 + r_{\text{pole}}^{-2} k^2 > 0$$

for every nonzero decomposable bivector.

Equip the star quotient D_S with its induced metric g_S . The action is free and isometric. For horizontal lifts of tangent vectors, the submersion formula [11] reads

$$(4.6) \quad \mathcal{K}_{g_S}(X \wedge Y) = \mathcal{K}_{G_S}(\tilde{X} \wedge \tilde{Y}) + \frac{3}{4} |[\tilde{X}, \tilde{Y}]^{\text{V}^*}|^2.$$

Here the lifts are extended as basic horizontal fields for the star quotient, and $V_\star$ denotes orthogonal projection onto the star-vertical space $\ker d\varpi_S$. Thus D_S is a smooth positively curved compact disk, including its boundary and center.

4.3. The boundary data. The connection potential A_N vanishes on a whole boundary collar. Thus (4.1) is a doubly warped product in that gauge. Evaluating the radii and their logarithmic derivatives at $t = a$ gives (4.2) and the boundary metric in Theorem 4.1. The outward normal is $-\partial_t$, so differentiating the two warped factors gives (4.3). The ratio q_s/r_a^2 tends to $A_0 F_a/2$ as $\varepsilon \rightarrow 0$. The northern angular profile will be chosen in terms of A_0 and F_a to accommodate this boundary slope.

Proof of Theorem 4.1. The connection constructed above is smooth and star invariant. The height function $\varphi = -A_0 \cos t$ satisfies the hypotheses of Proposition 3.1, so the choice of ε_S gives the asserted source curvature bound. The pole calculation and O'Neill's formula (4.6) give smoothness and the same positive lower bound on the quotient. Its disk identification is the one in Section 2. The boundary calculation above proves the remaining assertions. $\square$

5. THE NORTHERN FILLING

We now construct a filling with the boundary radii and fibre slope provided by Theorem 4.1. Fix $a \in (\pi/2, \pi)$ and $A_0 > 0$, and define F_a, r_a, q_s for each $\varepsilon > 0$ by (4.2). In the product trivialization $P_N \cong D^7 \times S^3$, consider

$$(5.1) \quad G_N = ds^2 + F(s)^2 h_{S^6} + r(s)^2 h_{S^3}, \quad 0 \leq s \leq \ell_N.$$

The angular boundary coordinates are those of Section 2; the connection is flat, and the star action is (2.4).

Theorem 5.1 (Northern filling with prescribed boundary data). *There are $\ell_N > 0$, $\varepsilon_N > 0$ and $F \in C^\infty([0, \ell_N])$, chosen independently of ε , such that for every $0 < \varepsilon < \varepsilon_N$ there is a positive smooth function $r = r_\varepsilon$ on $[0, \ell_N]$ with the following properties. The function F is positive on $(0, \ell_N]$, extends to an odd smooth function near zero, and satisfies $F'(0) = 1$; the function r is constant near zero. The metric (5.1) therefore extends smoothly over $D^7 \times S^3$. Its star action is free and isometric, and every star-horizontal two-plane has strictly positive sectional curvature. The quotient (D_N, g_N) is a smooth marked seven-disk with $\sec_{g_N} > 0$, including at its center and boundary.*

The endpoint values satisfy

$$(5.2) \quad F(\ell_N) = F_a, \quad F'(\ell_N) > 0, \quad r(\ell_N) = r_a, \quad \frac{r'(\ell_N)}{r_a} = q_s.$$

Thus the source boundary metric is $F_a^2 h_{S^6} + r_a^2 h_{S^3}$. For a source boundary tangent vector $Z = X + U$ in orthonormal angular and fibre components, its outward second fundamental form $\mathcal{B}_N$ is

$$(5.3) \quad \mathcal{B}_N(Z, Z) = \mu_N |X|^2 + q_s |U|^2, \quad \mu_N := \frac{F'(\ell_N)}{F_a} > 0.$$

In particular, μ_N is independent of ε .

We first obtain a curvature criterion for the star-horizontal planes, then choose the two warping functions and check the center. The resulting quotient also satisfies the explicit lower bound (5.18).

5.1. Horizontal planes and the curvature formula. At $x = (p, w) \in S^6$, let

$$K_x : \text{Im } \mathbb{H} \longrightarrow T_x S^6$$

be the infinitesimal conjugation action. Splitting w into its imaginary and real parts, in that order, gives

$$K_x \xi = 2(\xi \times p, \xi \times \text{Im } w, 0), \quad \|K_x\|_{\text{op}} \leq 2.$$

The bound follows from $|p|^2 + |\text{Im } w|^2 \leq 1$. We write K_x^* for the adjoint with respect to Q and the unit round metric h_{S^6} .

Principal-right translation is an isometry commuting with the star action, so it suffices to work at $u = 1$. In orthonormal angular and fibre components a star generator is $(FK_x \xi, r\xi)$. A star-horizontal vector is therefore of the form

$$(5.4) \quad \lambda \partial_s + X + U, \quad U = TX, \quad T = -\frac{F}{r} K_x^*, \quad \|T\|_{\text{op}} \leq \frac{2F}{r}.$$

Principal-right translation transports this description isometrically to every point of the fibre.

For a second horizontal vector $\mu \partial_s + Y + V$, with $V = TY$, we have

$$(5.5) \quad |\lambda V - \mu U|^2 \leq \frac{4F^2}{r^2} |\lambda Y - \mu X|^2,$$

$$(5.6) \quad |X \otimes V - Y \otimes U|^2 \leq \frac{8F^2}{r^2} |X \wedge Y|^2.$$

The first inequality is the operator-norm estimate for T . To prove the second when X, Y are independent, replace the pair by an area-preserving linear change for which they are orthogonal. The bivector and the tensor on the left are unchanged. In the new pair the left side is

$$|X|^2 |TY|^2 + |Y|^2 |TX|^2 \leq 2 \|T\|_{\text{op}}^2 |X|^2 |Y|^2.$$

For linearly dependent X, Y , both sides of (5.6) vanish. These estimates express the vertical area terms in terms of the projected base areas.

For arbitrary vectors as above in the doubly warped source, the full sectional numerator is

$$(5.7) \quad \begin{aligned} \mathcal{K} = & -\frac{F''}{F} |\lambda Y - \mu X|^2 - \frac{r''}{r} |\lambda V - \mu U|^2 + \frac{1 - F'^2}{F^2} |X \wedge Y|^2 \\ & + \frac{1 - r'^2}{r^2} |U \wedge V|^2 - \frac{F' r'}{F r} |X \otimes V - Y \otimes U|^2. \end{aligned}$$

To obtain this formula, take orthonormal frames in the two round factors and evaluate at a point where their intrinsic frames are normal:

$$\begin{aligned} \nabla_{\partial_s} E_i &= \nabla_{\partial_s} E_a = 0, & \nabla_{E_i} \partial_s &= (F'/F) E_i, & \nabla_{E_a} \partial_s &= (r'/r) E_a, \\ \nabla_{E_i} E_j &= -(F'/F) \delta_{ij} \partial_s, & \nabla_{E_a} E_b &= -(r'/r) \delta_{ab} \partial_s, & \nabla_{E_i} E_a &= \nabla_{E_a} E_i = 0. \end{aligned}$$

Differentiating the connection formulas and including the intrinsic curvature tensors of the two round factors gives a curvature operator that is diagonal across the radial–first-factor, radial–second-factor, first-factor, second-factor and mixed-factor bivector blocks. Their coefficients are exactly the five coefficients in (5.7).

Lemma 5.2 (Horizontal curvature criterion). *Suppose $F, r > 0$, $F', r' \geq 0$, and*

$$(5.8) \quad \frac{1 - F'^2}{F^2} > \frac{8FF'r'}{r^3}, \quad -\frac{F''}{F} > \frac{4F^2(r'')_+}{r^3}, \quad r' < 1.$$

Then every star-horizontal two-plane has positive sectional curvature.

Proof. Insert (5.5) and (5.6) into (5.7). The possible negative radial-fibre term is bounded using $(r'')_+ = \max\{r'', 0\}$. The inequalities leave strictly positive coefficients of

$$|\lambda Y - \mu X|^2, \quad |X \wedge Y|^2,$$

and a nonnegative remaining contribution from $|U \wedge V|^2$. The first two areas cannot both vanish on an independent horizontal pair. Indeed, projection of the graph (5.4) to $\mathbb{R}\partial_s \oplus T_x S^6$ is injective; their simultaneous vanishing would make its two projected vectors, and hence the original two vectors, dependent. This proves the lemma. $\square$

Remark 5.3 (The two horizontal distributions). The principal-horizontal space of (5.1) is tangent to the base disk. The star-horizontal space is the graph (5.4). On this graph, (5.5) and (5.6) bound the radial-fibre and mixed angular contributions by base-area terms. Lemma 5.2 therefore allows r'' to be positive. This accommodates a radius that is constant near the center and satisfies $r'(\ell_N) = r_a q_s > 0$ at the boundary.

5.2. Choice of the warping functions. The prescribed positive boundary fibre slope explains the different roles of F and r . A radius that is constant near the center and has positive derivative at the boundary cannot be concave throughout the disk. On star-horizontal planes, the resulting radial loss is bounded by $4F^2(r'')_+/r^3$, whereas the angular loss is bounded by $8FF'r'/r^3$. Small fibres alone do not make the latter harmless: at the boundary,

$$\frac{r'(\ell_N)}{r(\ell_N)^3} = \frac{q_s}{r_a^2} \longrightarrow \frac{A_0 F_a}{2} \quad (\varepsilon \downarrow 0).$$

We therefore choose the angular geometry before the common fibre scale.

The profile (5.10) provides radial curvature of order δ^{-2} where the fibre slope changes, and makes F' decay rapidly as F/δ grows. We confine the change of fibre slope to $\delta/4 \leq s \leq \delta/2$. On this interval the radial loss is of order at most $A_0 F_a C_\eta \delta$; outside it $r'' = 0$, so this loss vanishes. For the angular coefficient, the useful ratio is

$$\frac{(1 - F'^2)/F^2}{FF'} = \delta^{-3} \frac{1 - e^{-x^2}}{x^3 e^{-x^2/2}}, \quad x = F/\delta.$$

The positive lower bound in (5.17) shows why choosing $\delta^3 A_0 F_a$ small controls the angular loss on the whole disk. The estimates below make both comparisons quantitative. Although this profile can produce a long interval, its length is finite and fixed before ε is chosen.

Fix a smooth nondecreasing function $\eta : \mathbb{R} \rightarrow [0, 1]$ with

$$\eta = 0 \text{ on } (-\infty, 1/4], \quad \eta = 1 \text{ on } [1/2, \infty).$$

Let $C_\eta = \sup |\eta'|$. For the fixed A_0, F_a , choose

$$(5.9) \quad 0 < \delta \leq F_a/2, \quad \delta^3 \leq \frac{1}{128 A_0 F_a (1 + C_\eta)}.$$

Define F by

$$(5.10) \quad F' = e^{-F^2/(2\delta^2)}, \quad F(0) = 0, \quad \ell_N = \int_0^{F_a} e^{z^2/(2\delta^2)} dz.$$

Equivalently, $s = \int_0^{F(s)} e^{z^2/(2\delta^2)} dz$. This defines a smooth increasing solution with $F(\ell_N) = F_a$. Both F and the finite length ℓ_N are determined by F_a, δ .

For the values r_a, q_s in (4.2), put

$$(5.11) \quad d = r_a q_s, \quad r(s) = r_a - d \int_s^{\ell_N} \eta(v/\delta) dv.$$

For the northern filling, require

$$(5.12) \quad q_s \ell_N \leq \frac{1}{2}, \quad d < 1.$$

These are upper bounds satisfied for all sufficiently small $\varepsilon > 0$, since $q_s = O(\varepsilon)$ and $d = O(\varepsilon^{3/2})$, with all other parameters fixed. We obtain

$$(5.13) \quad \frac{r_a}{2} \leq r \leq r_a, \quad r' = d\eta(s/\delta) \geq 0, \quad r'' = \frac{d}{\delta}\eta'(s/\delta) \geq 0.$$

Since $\ell_N \geq F_a \geq 2\delta$, the function $\eta(s/\delta)$ is one near the outer boundary. Consequently

$$(5.14) \quad r(\ell_N) = r_a, \quad \frac{r'(\ell_N)}{r_a} = q_s.$$

We verify (5.8) on the whole interval $(0, \ell_N]$. First, (5.13) gives

$$(5.15) \quad \frac{d}{r^3} \leq \frac{8q_s}{r_a^2} = 4A_0 F_a e^{-\varepsilon A_0 |\cos a|} \leq 4A_0 F_a.$$

Differentiating (5.10) gives

$$F'' = -\frac{F}{\delta^2}(F')^2.$$

Thus

$$(5.16) \quad -\frac{F''}{F} = \delta^{-2} e^{-F^2/\delta^2}, \quad \frac{1 - F'^2}{F^2} = \frac{1 - e^{-F^2/\delta^2}}{F^2}.$$

The support of r'' is contained in $s \leq \delta/2$. Since $F' \leq 1$, we have $F(s) \leq s$, and hence there

$$-\frac{F''}{F} \geq e^{-1/4}\delta^{-2}, \quad \frac{4F^2 r''}{r^3} \leq 4A_0 F_a C_\eta \delta \leq \frac{C_\eta}{32(1 + C_\eta)} \delta^{-2}.$$

This proves the radial inequality. Off the support of r'' its right side vanishes and its left side remains strictly positive.

For the angular inequality, for every $x > 0$ we have

$$(5.17) \quad \frac{1 - e^{-x^2}}{x^3 e^{-x^2/2}} = \frac{2 \sinh(x^2/2)}{x^3} \geq \frac{1}{x} + \frac{x^3}{24} \geq \frac{4}{3 \cdot 8^{1/4}} > \frac{1}{2}.$$

The first bound follows from the first two nonzero terms in the power series of $\sinh$, whose subsequent terms are positive. The minimum of $x^{-1} + x^3/24$ occurs at $x^4 = 8$, giving the second bound. Substitution of $x = F/\delta$ gives

$$\frac{1 - F'^2}{F^2} > \frac{FF'}{2\delta^3} > 32A_0 F_a FF' \geq \frac{8FF'r'}{r^3},$$

where we used (5.9), (5.15), and $r' \leq d$. Finally, $r' \leq d < 1$, so all three inequalities in (5.8) hold.

5.3. The center and the quotient. The solution F extends to an odd smooth function near zero, with $F'(0) = 1$. Indeed, it is the local inverse of the odd smooth function $\int_0^F e^{z^2/(2\delta^2)} dz$, whose derivative at zero is one. These are the polar smoothness conditions for $ds^2 + F^2 h_{S^6}$. More explicitly,

$$F(s) = s - \frac{s^3}{6\delta^2} + O(s^5).$$

The function r is constant near zero by the choice of η , so the entire source metric is smooth there.

Both expressions in (5.16) tend to $\delta^{-2} > 0$ at the center. In a neighborhood of the center the source is a product of this positively curved base with a round S^3 of constant radius. Its sectional numerator on star-horizontal planes is positive, since projection of such a plane to the base is injective. At the center itself the star orbit is entirely in the fibre and the horizontal space is the whole base tangent space. Thus positivity also holds at that point.

Lemma 5.2 now proves positivity on all star-horizontal planes throughout the source, including the boundary. By (4.6), the star quotient D_N is a smooth positively curved compact disk. The estimates use the bound $\|K_x\|_{\text{op}} \leq 2$, which holds on all of S^6 , including points where the conjugation orbit has smaller dimension. More explicitly, at $u = 1$ the Gram matrix of the star infinitesimal generators is

$$F^2 K_x^* K_x + r^2 \text{Id}_{\text{Im } \mathbb{H}} \geq r^2 \text{Id}_{\text{Im } \mathbb{H}} > 0.$$

Thus a change in the rank of K_x does not make the star orbit singular. The horizontal graph formula uses K_x^* , not an inverse of K_x , and remains valid at every orbit type.

5.4. A quantitative lower bound on the quotient. We record a lower bound in terms of the chosen parameters. This also quantifies the passage from the horizontal curvature estimate to the quotient metric. Put

$$T_* := F_a^2/\delta^2 \geq 4, \quad p_* := \delta^{-2} e^{-T_*}, \quad q_* := \frac{1 - e^{-T_*}}{2F_a^2}.$$

Let $P_{\text{rad}}, P_{\text{ang}}$ be the two remaining base-area coefficients after applying (5.5) and (5.6):

$$P_{\text{rad}} = -F''/F - 4F^2(r'')_+/r^3, \quad P_{\text{ang}} = (1 - F'^2)/F^2 - 8FF'r'/r^3.$$

On the support of r'' , the radial estimates above give $P_{\text{rad}} > 15/(32\delta^2) > p_*$; off this support, $P_{\text{rad}} = \delta^{-2} e^{-F^2/\delta^2} \geq p_*$. For the angular term, (5.9), (5.15) and (5.17) imply

$$\frac{8FF'r'}{r^3} \leq \frac{FF'}{4(1 + C_\eta)\delta^3} < \frac{1}{2} \frac{1 - F'^2}{F^2}.$$

The function $(1 - e^{-y})/y$ is decreasing for $y > 0$, since $(y + 1)e^{-y} < 1$. It follows that

$$P_{\text{ang}} \geq \frac{1}{2} \frac{1 - e^{-F^2/\delta^2}}{F^2} \geq q_*.$$

All expressions at the center are interpreted by their smooth limits.

The horizontal graph injection in (5.4) has squared operator norm at most

$$H_* := 1 + 16F_a^2/r_a^2,$$

since $F \leq F_a$ and $r \geq r_a/2$. Its induced map on second exterior powers has squared operator norm at most H_*^2 . Consequently, for any unit star-horizontal bivector, the sum of its two

base-area squares is at least H_*^{-2} . The vertical-area term in (5.7) is nonnegative. O'Neill's formula therefore gives

$$(5.18) \quad \sec_{g_N} \geq c_N := \left(1 + \frac{16F_a^2}{r_a^2}\right)^{-2} \min \left\{ \frac{e^{-F_a^2/\delta^2}}{\delta^2}, \frac{1 - e^{-F_a^2/\delta^2}}{2F_a^2} \right\} > 0.$$

For fixed parameters this bound is positive, and it extends to the center by continuity.

Proof of Theorem 5.1. Choose η, δ as above and define F, ℓ_N by (5.10). These choices depend on A_0, F_a , not on ε . Choose $\varepsilon_N > 0$ so that (5.12) holds for every $0 < \varepsilon < \varepsilon_N$, and define r by (5.11). Equations (5.13) and (5.14) give its positivity, its constant value near the center, and its prescribed endpoint value and derivative. The preceding estimates verify Lemma 5.2 on $(0, \ell_N]$; the center calculation proves the smooth extension and positivity there. The free star quotient is a marked disk by Section 2, and O'Neill's formula gives (5.18). Finally, (5.10) and (5.14) yield (5.2). Differentiating (5.1) in the outward direction ∂_s gives (5.3). $\square$

6. BOUNDARY COMPATIBILITY AND GLUING

The two filling theorems give metrics with the same source boundary radii and fibre slope. We now compare their quotient boundaries under the prescribed marked map. Denote the induced boundary metrics by h_N, h_S and the outward second fundamental forms by B_N, B_S .

Proposition 6.1 (Boundary compatibility). *Fix $n \in \mathbb{Z}$ and the data in Theorems 4.1 and 5.1, using the same a, A_0 . For any $0 < \varepsilon < \min\{\varepsilon_S, \varepsilon_N\}$, let (D_S, g_S) and (D_N, g_N) be the resulting fillings. The marked map $\phi_n : \partial D_N \rightarrow \partial D_S$, represented by σ^n in the extending disk coordinates, is a boundary isometry:*

$$h_N = \phi_n^* h_S =: h.$$

With $\mu_N = F'(\ell_N)/F_a > 0$ and $\mu_S = \cot a < 0$,

$$(6.1) \quad B_N + \phi_n^* B_S \geq c_B h, \quad c_B := \frac{r_a^2}{r_a^2 + 4F_a^2} (\mu_N - \mu_S) > 0.$$

We establish the metric equality in the common north gauge and then compute the second fundamental forms on the same horizontal lifts.

6.1. The common boundary metric. In the north gauge the two source boundary metrics are both

$$(6.2) \quad F_a^2 h_{S^6} + r_a^2 h_{S^3}.$$

The star actions agree, so the induced quotient boundary metrics are exactly equal under the identity in that gauge. Write $\varpi_\partial(x, u) = u^{-1}xu$ for the boundary quotient map in the common angular marking. At $u = 1$, its differential in orthonormal components is

$$d\varpi_\partial(X, U) = F_a^{-1}X - r_a^{-1}K_x U.$$

The source inner product in these components is the identity. Hence, for $L = (F_a^{-1} \text{Id}, -r_a^{-1}K_x)$, the quotient cometric is LL^* , namely

$$(6.3) \quad h^{-1} = F_a^{-2} h_{S^6}^{-1} + r_a^{-2} K_x K_x^*.$$

By Remark 2.2, the identity in this trivialization is represented by σ^n in the extending disk coordinates.

6.2. The second fundamental forms. Denote the boundary isometry by $\phi_n : \partial D_N \rightarrow \partial D_S$. All comparisons below are on ∂D_N , with the southern tensors pulled back by ϕ_n . Let $Y \neq 0$ be a tangent vector to this boundary, and write its common horizontal lift as $X + U$. The outward unit normal is ∂_s on the north and $-\partial_t$ on the south. These normals are horizontal for the star submersions. The horizontal component of the covariant derivative of a basic horizontal normal projects to the quotient covariant derivative. Taking its inner product with the horizontal lift of a boundary vector identifies the quotient second fundamental form with the source form restricted to horizontal lifts. Since $A_N = 0$ on the boundary collar, the source forms follow by differentiating the two warping factors. With

$$\mu_N = \frac{F'(\ell_N)}{F_a} > 0, \quad \mu_S = \cot a < 0,$$

equation (5.14) gives

$$(6.4) \quad \begin{aligned} B_N(Y, Y) &= \mu_N |X|^2 + q_s |U|^2, \\ (\phi_n^* B_S)(Y, Y) &= -\mu_S |X|^2 - q_s |U|^2. \end{aligned}$$

Their sum is therefore

$$(6.5) \quad (B_N + \phi_n^* B_S)(Y, Y) = (\mu_N - \mu_S) |X|^2 > 0.$$

By (5.4), a nonzero horizontal vector has $X \neq 0$. Together with $\mu_N > 0 > \mu_S$, this proves strict positive definiteness.

To express the eigenvalues relative to h , let b_j be the eigenvalues of $K_x K_x^*$, so $0 \leq b_j \leq 4$. For X in a corresponding eigendirection,

$$|Y|_h^2 = |X|^2 + |U|^2 = \left(1 + \frac{F_a^2 b_j}{r_a^2}\right) |X|^2.$$

The eigenvalues of the shape sum relative to h are thus

$$(6.6) \quad \frac{r_a^2}{r_a^2 + F_a^2 b_j} (\mu_N - \mu_S) > 0.$$

Since $b_j \leq 4$, these eigenvalues are bounded below by the constant c_B in (6.1). At $x = (i, 0)$, the map $K_x K_x^*$ has eigenvalue 4, so c_B is attained. On $\ker K_x^*$, the eigenvalue of the shape sum is $\mu_N - \mu_S = \mu_N + |\cot a|$.

Proof of Proposition 6.1. Theorems 4.1 and 5.1 give the common source boundary metric (6.2). Taking the quotient by the common star action gives the boundary isometry above, whose expression in the extending coordinates is σ^n by (2.5). The horizontal restriction of the source forms (4.3) and (5.3) gives (6.4). Their sum and its eigenvalues are (6.5) and (6.6), proving (6.1). $\square$

6.3. The compatible filling theorem.

Proof of Theorem 1.3. Fix $n \in \mathbb{Z}$. Choose a, b_0, χ and the southern connection, which determines M_0, M_1 . Choose Λ, A_0 as in Theorem 4.1; this theorem supplies a family of southern fillings for $0 < \varepsilon < \varepsilon_S$. With this same a, A_0 , Theorem 5.1 supplies F, ℓ_N and a family of northern fillings for $0 < \varepsilon < \varepsilon_N$. The northern angular profile is fixed before ε , so both thresholds are determined before making the common choice

$$0 < \varepsilon < \min\{\varepsilon_S, \varepsilon_N\}.$$

In particular, ε satisfies (3.18)–(3.20) and (5.12) for the fixed data.

Theorems 4.1 and 5.1 give smooth positive metrics on the entire marked disks for this same ε . Proposition 6.1 gives $h_N = \phi_n^* h_S$ and $B_N + \phi_n^* B_S > 0$ under their prescribed marked identification. These are precisely the conclusions of Theorem 1.3. $\square$

6.4. Positive gluing and the main theorem. The $k = 1$ case of the Reiser–Wraith gluing theorem [14, Theorem A(i)] concerns positive sectional curvature. It assumes that the boundary metrics are isometric and that the sum of the second fundamental forms is positive semidefinite. We use the following strict case. With inward normal $N = -n_{\text{out}}$, our convention is $B(Y, Z) = -\langle \nabla_Y N, Z \rangle$, which agrees with the convention in [14].

Lemma 6.2 (Positive gluing). *Let (M_N, g_N) and (M_S, g_S) be smooth Riemannian manifolds of the same dimension at least two, with compact boundaries and strictly positive sectional curvature, including at the boundaries. Let h_N, h_S and B_N, B_S be their induced boundary metrics and outward second fundamental forms, with convention (1.2). Suppose a boundary diffeomorphism $\phi : \partial M_N \rightarrow \partial M_S$ satisfies*

$$h_N = \phi^* h_S =: h, \quad B_N + \phi^* B_S \geq b_* h \quad \text{for some } b_* > 0.$$

Then the glued smooth manifold $M_N \cup_\phi M_S$ admits a smooth metric of strictly positive sectional curvature. For every neighborhood of the identified boundary, the metric may be chosen to agree with g_N and g_S outside that neighborhood.

Proof. See Appendix B. $\square$

For the quotient disks, the signed normal coordinate is

$$z = s - \ell_N \leq 0 \quad \text{on } D_N, \quad z = t - a \geq 0 \quad \text{on } D_S.$$

The radial fields are unit and star-horizontal, and the star orbits are tangent to the radial slices. After pulling back the southern collar by ϕ_n , the quotient metrics therefore have the form $dz^2 + h_-(z)$ and $dz^2 + h_+(z)$, with

$$h_-(0) = h_+(0) = h, \quad h'_-(0) - h'_+(0) = 2(B_N + \phi_n^* B_S) \geq 2c_B h.$$

The fibre radii have the common value r_a and common derivative $r_a q_s$ at $z = 0$. The angular derivatives give the positive jump displayed above. Thus the two metrics define a continuous metric with a corner along the identified boundary.

Fix the common ε as in the proof of Theorem 1.3. The curvature bounds c_S, c_N and the boundary constant c_B are then positive fixed numbers. The interpolation width τ is chosen for these metrics, and the smoothing width ν is chosen after τ .

Proof of Theorem 1.1. Choose an orientation on the given homotopy seven-sphere Σ . By the coordinate identification in Lemma 2.1 and [3, Theorem 1], with the orientation comparison in Section 2, its oriented smooth type is represented by Σ_n for some $n \in \{0, \dots, 27\}$. Theorem 1.3 provides smooth positive metrics on the two marked disks defining Σ_n , with isometric boundaries and a positive definite sum of outward second fundamental forms. Their boundaries are compact, so Lemma 6.2 applies and gives a smooth positively curved metric on $D_N \cup_{\phi_n} D_S = \Sigma_n$. The quantitative version in [14, Corollary B, $k = 1$] applies with lower threshold $\frac{1}{2} \min\{c_S, c_N\}$, since both disk metrics have strictly larger sectional curvature. Thus the smoothed metric may be chosen to satisfy

$$\sec_g > \frac{1}{2} \min\{c_S, c_N\} > 0.$$

Pulling this metric back to Σ proves the theorem, with $c_\Sigma = \frac{1}{2} \min\{c_S, c_N\}$. $\square$

APPENDIX A. THE PRINCIPAL BUNDLE P_n

We give the quotient-space construction underlying the clutching description in Section 2, using the notation fixed there.

For each $n \in \mathbb{Z}$, the principal right S^3 -bundle

$$\pi_n : P_n \longrightarrow S^7$$

has structure group S^3 , the group of unit quaternions under multiplication. Its total space is

$$P_n := \left((\mathcal{U}_N \times S^3) \sqcup (\mathcal{U}_S \times S^3) \right) / \sim_n,$$

where $\sqcup$ denotes disjoint union. Write $(y, u)_\alpha$ for a point in the α -th product, with $\alpha \in \{N, S\}$. The equivalence relation $\sim_n$ is generated by identifying $(y, u_N)_N$ with $(y, u_S)_S$ whenever the common base point is $y = (\cos t, \sin t x) \in \mathcal{U}_N \cap \mathcal{U}_S$ and

$$(A.1) \quad u_S = g_n(x)u_N.$$

Equip P_n with the quotient topology and denote these classes by $[y, u]_\alpha$. Since the identifications preserve the base point, the product projections induce a well-defined continuous map

$$\pi_n : P_n \longrightarrow S^7, \quad \pi_n([y, u]_\alpha) = y.$$

The canonical maps

$$\iota_\alpha : \mathcal{U}_\alpha \times S^3 \longrightarrow \pi_n^{-1}(\mathcal{U}_\alpha), \quad \iota_\alpha(y, u) = [y, u]_\alpha, \quad \alpha \in \{N, S\},$$

are homeomorphisms onto open subsets covering P_n . The quotient is Hausdorff: points with different projections are separated using the base, and points with the same projection are separated in a common product trivialization. It is second countable because these two open subsets are second countable. On the overlap,

$$(\iota_S^{-1} \circ \iota_N)(y, u) = (y, g_n(x)u), \quad y = (\cos t, \sin t x).$$

This transition is a smooth diffeomorphism, with inverse $(y, u) \mapsto (y, g_n(x)^{-1}u)$. We give P_n the unique smooth structure for which both ι_N and ι_S are diffeomorphisms. Thus P_n is a smooth ten-dimensional manifold without boundary, and π_n is a smooth locally trivial bundle with fibre S^3 .

Define the *principal right action* by

$$[y, u]_\alpha \cdot h := [y, uh]_\alpha, \quad h \in S^3.$$

The identity $g_n(x)(uh) = (g_n(x)u)h$ shows that this definition is independent of the representative. In each trivialization it is the standard smooth right multiplication on S^3 ; hence it preserves π_n and is free and transitive on every fibre. Consequently, $\pi_n : P_n \rightarrow S^7$, with these trivializations and this action, is a smooth principal right S^3 -bundle.

APPENDIX B. PROOF OF THE POSITIVE GLUING LEMMA

We prove Lemma 6.2 by the cubic interpolation of Reiser–Wraith [14, Section 2], followed by smoothing in normal collars. Throughout the proof, the two metrics are fixed. The interpolation width is denoted by $\tau > 0$.

B.1. Signed collars and the boundary inequality. If the common boundary is empty, the disjoint union of the two original metrics already proves the lemma. Assume henceforth that it is nonempty. Set $X = \partial M_N$, and use ϕ to pull the southern collar back to X . In signed normal coordinates, increasing from the northern side to the southern side, the metrics take the form

$$(B.1) \quad g_N = dz^2 + h_-(z) \quad (-r_0 \leq z \leq 0), \quad g_S = dz^2 + h_+(z) \quad (0 \leq z \leq r_0),$$

where $h_-(0) = h_+(0) = h$. Here and below the notation g_S on the collar includes this pullback. The families $h_\pm$ are smooth up to $z = 0$. Compactness of X allows a common $r_0 > 0$; the collars may be chosen inside any prescribed open neighborhood of the identified boundary. These collars define the usual smooth structure on the gluing by ϕ .

The outward normals are $+\partial_z$ on the northern side and $-\partial_z$ on the southern side. Consequently

$$(B.2) \quad P_0 := h'_-(0), \quad Q_0 := h'_+(0), \quad \Delta := P_0 - Q_0 = 2(B_N + \phi^* B_S) \geq 2b_* h.$$

Primes denote z -derivatives. All tensor estimates below are uniform on X , measured using h and a finite collection of fixed coordinate charts. An estimate in C_X^j includes all tangential derivatives of order at most j , but not normal derivatives. The latter will be stated separately.

B.2. Cubic interpolation and its derivatives. For $0 < \tau < r_0/4$, put

$$a_\tau = h_-(-\tau), \quad b_\tau = h_+(\tau), \quad P_\tau = h'_-(-\tau), \quad Q_\tau = h'_+(\tau), \quad D_\tau = \frac{b_\tau - a_\tau}{2\tau}.$$

On $-\tau \leq z \leq \tau$, define

$$(B.3) \quad H_z = \frac{z + \tau}{2\tau} b_\tau - \frac{z - \tau}{2\tau} a_\tau + \frac{(z - \tau)^2(z + \tau)}{4\tau^2} (P_\tau - D_\tau) + \frac{(z + \tau)^2(z - \tau)}{4\tau^2} (Q_\tau - D_\tau).$$

This is a polynomial with values in $C^\infty(X, \text{Sym}^2 T^*X)$. Its endpoint data are

$$(B.4) \quad H_{-\tau} = a_\tau, \quad H_\tau = b_\tau, \quad H'_{-\tau} = P_\tau, \quad H'_\tau = Q_\tau.$$

Differentiating the polynomial gives

$$(B.5) \quad H'_z = D_\tau + \frac{3z^2 - \tau^2}{4\tau^2} (P_\tau + Q_\tau - 2D_\tau) + \frac{z}{2\tau} (Q_\tau - P_\tau),$$

$$(B.6) \quad H''_z = \frac{3z}{2\tau^2} (P_\tau + Q_\tau - 2D_\tau) + \frac{Q_\tau - P_\tau}{2\tau}.$$

Taylor expansion of the fixed smooth families yields, in every fixed tangential C^j -norm,

$$a_\tau = h - \tau P_0 + O(\tau^2), \quad b_\tau = h + \tau Q_0 + O(\tau^2),$$

$$P_\tau = P_0 + O(\tau), \quad Q_\tau = Q_0 + O(\tau), \quad D_\tau = \frac{1}{2}(P_0 + Q_0) + O(\tau).$$

In particular, $P_\tau + Q_\tau - 2D_\tau = O(\tau)$ and $Q_\tau - P_\tau = -\Delta + O(\tau)$. Substitution into (B.3)–(B.6), with $\lambda = (z + \tau)/(2\tau)$, gives the uniform estimates

$$(B.7) \quad H_z = h + O(\tau) \quad \text{in } C_X^2,$$

$$(B.8) \quad H'_z = (1 - \lambda)P_0 + \lambda Q_0 + O(\tau) \quad \text{in } C_X^1,$$

$$(B.9) \quad H''_z = -\frac{\Delta}{2\tau} + O(1) \quad \text{in } C_X^0.$$

For example, the first term on the right side of (B.6) is $O(1)$, since $|z| \leq \tau$ and its bracket is $O(\tau)$. To obtain (B.8), the $O(1)$ part is $(P_0 + Q_0)/2 + z(Q_0 - P_0)/(2\tau)$, which is the stated linear interpolation. The same calculations commute with tangential differentiation because all scalar coefficients depend only on z .

After decreasing τ , (B.7) ensures $\frac{1}{2}h \leq H_z \leq 2h$. Thus $G_\tau = dz^2 + H_z$ is a Riemannian metric on the interpolation strip. Extend it by g_N and g_S outside this strip. Equation (B.4), as an identity of smooth tensors on X , matches both values and all first derivatives at the two interfaces $z = \pm\tau$. The extended metric is therefore C^1 and piecewise smooth. For a fixed τ , it is locally $C^{1,1}$, with bounded one-sided second coordinate derivatives. We next estimate the classical curvature on its smooth pieces and their one-sided limits.

B.3. All curvature blocks in a normal collar. Consider a smooth metric $G = dz^2 + H_z$. Let indices i, j, k, l be tangent to X , and let D^H be the Levi-Civita connection of the slice metric H_z . The Christoffel symbols in the product coordinates are

$$(B.10) \quad \Gamma_{ij}^z = -\frac{1}{2}H'_{ij}, \quad \Gamma_{zj}^i = \frac{1}{2}H^{ik}H'_{kj}, \quad \Gamma_{jk}^i = \Gamma_{jk}^i(H_z), \quad \Gamma_{zz}^A = \Gamma_{zi}^z = 0.$$

Here a capital index can be tangential or normal. Write $R_{ABCD} = G(R(\partial_A, \partial_B)\partial_C, \partial_D)$. Substituting (B.10) into the curvature definition gives

$$(B.11) \quad R_{ijkl} = R_{ijkl}^{H_z} + \frac{1}{4}(H'_{ik}H'_{jl} - H'_{il}H'_{jk}),$$

$$(B.12) \quad R_{izzj} = -\frac{1}{2}H''_{ij} + \frac{1}{4}H'_{ik}H^{kl}H'_{lj},$$

$$(B.13) \quad R_{ijkz} = \frac{1}{2}((D_j^H H')_{ik} - (D_i^H H')_{jk}).$$

For (B.11), the normal Christoffel products add the displayed quadratic term to the intrinsic curvature of the slice. For (B.12), differentiating $H^{ik}H'_{kj}/2$ and using $(H^{-1})' = -H^{-1}H'H^{-1}$ gives the stated expression after lowering the remaining index. For (B.13), the tangential derivatives of $-H'_{jk}/2$, together with the slice Christoffel terms, combine into the two covariant derivatives shown. These three blocks and the curvature symmetries describe the entire tensor: a nonzero component has zero, one, or two normal indices, with at most one normal index in either skew pair.

For a nonzero tangential vector u , the radial formula reads

$$(B.14) \quad \sec_G(u, \partial_z) = \frac{-\frac{1}{2}H''_z(u, u) + \frac{1}{4}H'_z(u, \cdot)H_z^{-1}H'_z(\cdot, u)}{H_z(u, u)}.$$

Here $\sec_G(u, v)$ means the sectional curvature of the plane spanned by u, v .

B.4. Tangential planes and the determinant identity. First suppose that the manifold dimension is at least three. For a two-plane $\Pi \subset T_x X$, choose an h -orthonormal basis u, v , extended with constant components in the product direction. If S is a symmetric two-tensor, write $\det_\Pi S = S(u, u)S(v, v) - S(u, v)^2$. The Gauss equation gives

$$(B.15) \quad \sec_{G_\tau}(\Pi) = \sec_{H_z}(\Pi) - \frac{\det_\Pi H'_z}{4 \det_\Pi H_z}.$$

Since $\det_\Pi h = 1$, (B.7) implies $\det_\Pi H_z = 1 + O(\tau)$ and $\sec_{H_z}(\Pi) = \sec_h(\Pi) + O(\tau)$, uniformly over the compact bundle of these planes. The normalization of the basis makes the denominator uniformly bounded away from zero.

For two symmetric 2×2 matrices P, Q , direct expansion of the determinant yields

$$(B.16) \quad \det((1 - \lambda)P + \lambda Q) = (1 - \lambda) \det P + \lambda \det Q - \lambda(1 - \lambda) \det(P - Q).$$

Apply this identity to the restrictions of P_0, Q_0 to Π . Their difference is the restriction of Δ , which is positive definite by (B.2), so $\det_\Pi \Delta > 0$. Equations (B.8) and (B.15) give

$$(B.17) \quad \begin{aligned} \sec_{G_\tau}(\Pi) &= (1 - \lambda) \sec_{g_N}(\Pi) + \lambda \sec_{g_S}(\Pi) \\ &\quad + \frac{\lambda(1 - \lambda)}{4} \det_\Pi \Delta + O(\tau). \end{aligned}$$

The curvatures on the right are evaluated at the boundary, with the southern plane transported by ϕ . Indeed, their Gauss formulas are $\sec_h(\Pi) - \frac{1}{4} \det_\Pi P_0$ and $\sec_h(\Pi) - \frac{1}{4} \det_\Pi Q_0$, respectively. The sign of the southern outward normal does not change this determinant.

Both boundary curvatures are strictly positive. Compactness therefore gives a common lower bound $\kappa_\partial > 0$ over all tangential two-planes. Decreasing τ once more in (B.17) yields

$$(B.18) \quad \sec_{G_\tau}(\Pi) \geq a_0 := \kappa_\partial/2 > 0 \quad (\Pi \subset T_x X).$$

The ambient boundary curvatures and the positive definite tensor Δ thus give the tangential lower bound.

B.5. Radial and arbitrary two-planes. For an H_z -unit tangential vector u , (B.9) and (B.14) imply

$$\sec_{G_\tau}(u, \partial_z) = \frac{\Delta(u, u)}{4\tau} + O(1).$$

The error is uniform: H'_z, H_z^{-1} , and the remainder in (B.9) are bounded independently of sufficiently small τ . Moreover, $H_z \leq 2h$ and $H_z(u, u) = 1$ give $h(u, u) \geq 1/2$, so $\Delta(u, u) \geq 2b_* h(u, u) \geq b_*$. There is a constant $C_0 \geq 0$ such that

$$(B.19) \quad \sec_{G_\tau}(u, \partial_z) \geq \frac{\beta_0}{\tau} - C_0, \quad \beta_0 := b_*/4 > 0.$$

Also, (B.7), (B.8), and (B.13) give a constant $M \geq 0$, independent of sufficiently small τ , such that

$$(B.20) \quad |G_\tau(R(u, w)w, \partial_z)| \leq M$$

whenever u, w are H_z -orthonormal. In fact, this block uses only H'_z , its first tangential derivatives, and the slice connection; all are uniformly bounded by the stated C_X^1 and C_X^2 estimates. It contains no H''_z .

Every two-plane in $\mathbb{R}\partial_z \oplus T_x X$ has a G_τ -orthonormal basis of the form

$$w, \quad \cos \theta u + \sin \theta \partial_z,$$

with $u, w \in T_x X$ orthonormal, when the manifold dimension is at least three. To see this, choose a unit vector in its intersection with $T_x X$, and orthogonally project the other basis vector onto $T_x X$. If that projection vanishes, choose any tangential unit vector orthogonal to w . For a tangential plane take $\theta = 0$. Expansion of the sectional numerator, followed by (B.18)–(B.20), gives

$$(B.21) \quad \begin{aligned} \sec_{G_\tau}(\Pi) &\geq a_0 \cos^2 \theta + \left(\frac{\beta_0}{\tau} - C_0 \right) \sin^2 \theta - 2M |\sin \theta \cos \theta| \\ &\geq \frac{a_0}{2} \cos^2 \theta + \left(\frac{\beta_0}{\tau} - C_0 - \frac{2M^2}{a_0} \right) \sin^2 \theta. \end{aligned}$$

Choose τ small enough that the coefficient in parentheses in the last line is at least $a_0/2$. Then every two-plane has sectional curvature at least $a_0/2$ on the interpolation strip. This bound is uniform in the angle; it includes planes whose angle depends on τ .

If the manifold dimension is two, the slice has dimension one and there are no tangential two-planes. The unique two-plane at each point is spanned by ∂_z and a slice unit vector. Its curvature is positive for small τ directly by (B.19). Thus the interpolation conclusion holds in that dimension as well.

Fix such a τ from now on. On the compact original subcollars the curvature has a positive minimum, and on the interpolation strip the preceding estimates give a positive minimum. Hence, on a compact collar containing both interfaces, there is a number $c_* > 0$ bounding below the sectional curvatures of all smooth pieces and their one-sided limits.

B.6. A common tensor-valued mollifier. We describe the smoothing near one interface, translated to $z = 0$. Write the fixed piecewise smooth metric as $G = dz^2 + k_z$. Choose $\ell > 0$ so small that $X \times [-3\ell, 3\ell]$ contains no other interface, is contained in the chosen collar, and is disjoint from the strip used at the other interface. All constants in the rest of the proof may depend on the fixed τ and on ℓ .

The tensor family k_z and its first normal derivative agree across zero as smooth tensors on X , by (B.4). The same is true after any number of tangential derivatives. In particular, in each product chart G is C^1 , its weak second derivatives are bounded, and they equal the ordinary second derivatives on the two open sides. There is no singular measure at the interface: the coefficient of a possible Dirac term in a second derivative would be a jump of a first derivative, and every such jump is zero.

Choose $\rho \in C_c^\infty((-1, 1))$ with $\rho \geq 0$ and $\int \rho = 1$. For $0 < \nu < \ell/2$, put $\rho_\nu(t) = \nu^{-1}\rho(t/\nu)$ and define

$$(B.22) \quad k_z^\nu(x) = \int \rho_\nu(z-s)k_s(x) ds, \quad G^\nu = dz^2 + k_z^\nu \quad (|z| \leq 2\ell).$$

Only $|s| < 3\ell$ contributes to this integral, so no extension outside the chosen collar is needed. The integration takes place in the single vector space $\text{Sym}^2 T_x^* X$, for fixed x . In particular the same nonnegative kernel is used for every tensor component. Product coordinate changes on X are independent of z , so this operation is intrinsically defined on the collar; no coordinate-dependent patching of separately smoothed components is needed. Positive definiteness is preserved by the integral. Tangential differentiation commutes with it, and normal differentiation acts on the smooth kernel. Thus G^ν is smooth in all variables. Uniform continuity of the first derivatives of G on the compact collar gives

$$(B.23) \quad \|G^\nu - G\|_{C^1(X \times [-2\ell, 2\ell])} \longrightarrow 0.$$

To control curvature, we estimate the linear second-derivative terms and the nonlinear first-derivative terms separately.

B.7. Curvature under the tensor-valued smoothing. In fixed product coordinates, set

$$\Gamma_{ABC}(G) = \frac{1}{2}(G_{BC,A} + G_{AC,B} - G_{AB,C}).$$

With all indices lowered, the curvature formula for a general positive definite metric is

$$(B.24) \quad \begin{aligned} R_{ABCD}(G) = & \frac{1}{2}(G_{BD,AC} - G_{BC,AD} - G_{AD,BC} + G_{AC,BD}) \\ & + G^{EF}(\Gamma_{ACE}(G)\Gamma_{BDF}(G) - \Gamma_{BCE}(G)\Gamma_{ADF}(G)). \end{aligned}$$

In particular,

$$(B.25) \quad R(G) = \mathcal{L}(D^2G) + \mathcal{Q}(G, DG),$$

where D here denotes coordinate differentiation, $\mathcal{L}$ is a constant-coefficient linear map of second coordinate derivatives, and $\mathcal{Q}$ is the smooth expression in the second line of (B.24). The latter involves no second derivative. Its dependence on G is smooth on the cone of positive definite matrices.

Apply (B.25) to G^ν . Convolution commutes with all coordinate derivatives in the weak sense. The absence of a singular measure in D^2G therefore gives the exact identity

$$(B.26) \quad R(G^\nu) = \rho_\nu * R(G) + E_\nu, \quad E_\nu = \mathcal{Q}(G^\nu, DG^\nu) - \rho_\nu * \mathcal{Q}(G, DG).$$

The curvature of G on the right is evaluated on its two smooth sides; its values on the interface, a set of measure zero in the convolution variable, are immaterial.

To quantify the error, work on a smaller fixed coordinate chart whose closure lies in a product chart. Uniform positive definiteness and (B.23) put the pairs (G^ν, DG^ν) and (G, DG) in a fixed compact subset of the domain of $\mathcal{Q}$. The mean value estimate and a modulus of continuity $\omega_{\mathcal{Q}}$ of the continuous function $\mathcal{Q}(G, DG)$ yield

$$(B.27) \quad \|E_\nu\|_{C^0} \leq C\|G^\nu - G\|_{C^1} + \omega_{\mathcal{Q}}(\nu) \longrightarrow 0.$$

The finitely many chart estimates give uniform convergence on the collar. Although the separate expressions in (B.25) are coordinate expressions, the difference E_ν in (B.26) is a tensor: all curvature tensors in the convolution are identified using the product bundle $TX \oplus \mathbb{R}\partial_z$, and the chart changes on this bundle are independent of z .

We now pass from (B.27) to a lower bound on every two-plane. Let $\bar{G} = dz^2 + h$, using the common boundary metric transported by the product structure. On the fixed strip choose constants $0 < m_* \leq M_* < \infty$ such that

$$(B.28) \quad m_*\bar{G} \leq G \leq M_*\bar{G}.$$

The same bounds hold for G^ν by (B.22). For a decomposable bivector $\xi \in \Lambda^2(T_x X \oplus \mathbb{R}\partial_z)$, fixed as z varies, the positive sectional-curvature bound on each smooth side gives

$$(B.29) \quad \mathcal{R}_{G(x,z)}(\xi, \xi) \geq c_*|\xi|_{G(x,z)}^2 \geq c_*m_*^2|\xi|_{\bar{G}}^2 \quad \text{for almost every } z.$$

Here $\mathcal{R}$ is the bilinear form on Λ^2 fixed in Section 1.3; in particular $\mathcal{R}_G(A \wedge B, A \wedge B) = G(R(A, B)B, A)$. The factor m_*^2 follows by taking the second exterior power of the metric comparison in (B.28).

Convolve (B.29) with the common nonnegative kernel. The product identification preserves decomposability, so the same bivector can be used in every term of the integral. The linear conversion of the all-lowered tensor R to $\mathcal{R}$ in (B.26), and (B.27), then give

$$\mathcal{R}_{G^\nu}(\xi, \xi) \geq (c_*m_*^2 - e_\nu)|\xi|_{\bar{G}}^2, \quad e_\nu \longrightarrow 0.$$

Here e_ν is an upper bound for the operator norm of the induced error bilinear form on Λ^2 , measured with $\bar{G}$. Choose ν so small that $e_\nu \leq c_*m_*^2/2$. Since $|\xi|_{G^\nu}^2 \leq M_*^2|\xi|_{\bar{G}}^2$, we obtain

$$(B.30) \quad \sec_{G^\nu} \geq \frac{c_*m_*^2}{2M_*^2} > 0 \quad (|z| \leq 2\ell).$$

The argument uses only decomposable bivectors, with the nonlinear curvature error E_ν controlled by (B.27).

B.8. Localization and completion of the proof. To leave the old metric unchanged away from the interface, choose $\zeta \in C_c^\infty((-2\ell, 2\ell))$ with $0 \leq \zeta \leq 1$ and $\zeta = 1$ on $[-\ell, \ell]$. Define

$$(B.31) \quad \widehat{k}_z = k_z + \zeta(z)(k_z^\nu - k_z), \quad \widehat{G} = dz^2 + \widehat{k}_z.$$

This is smooth at the interface because it equals G^ν there, and is smooth elsewhere because G is smooth away from the interface. It is positive definite, being a convex combination of the two positive definite slice metrics. Its curvature is positive where $\zeta = 1$ by (B.30), and where $\zeta = 0$ it is unchanged.

It remains to check the transition region $\ell \leq |z| \leq 2\ell$. For $\nu < \ell/2$, the convolution at these points samples only one smooth side of G . On this compact region, convolution therefore gives $k^\nu - k \rightarrow 0$ in C^2 , including all tangential and normal derivatives of total order at most two. Since ζ is fixed,

$$\|\widehat{G} - G\|_{C^2(\{\ell \leq |z| \leq 2\ell\})} \leq C_\zeta \|k^\nu - k\|_{C^2(\{\ell \leq |z| \leq 2\ell\})} \rightarrow 0.$$

The coordinate curvature formula and uniform positive definiteness imply convergence of sectional curvatures, uniformly on the compact bundle of two-planes in this region. The old metric has the positive lower bound c_* , so after decreasing ν again the localized metric is also strictly positively curved there. Thus the localized modification preserves positive sectional curvature by C^2 closeness on this smooth transition region.

Carry out this construction in the two disjoint strips about $z = -\tau$ and $z = \tau$, choosing the smoothing widths small enough for both. The metric is then smooth and strictly positively curved everywhere in the collar, and equals the original metrics outside the chosen neighborhood. This proves Lemma 6.2.

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